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NIDPS: Network Intrusion Detection and Prevention System

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Abstract

Our project presents a Network Intrusion Detection and Prevention System (NIDPS) designed to protect small networks using a combination of signature-based and machine learning techniques. The system uses Suricata to detect known attacks, Isolation Forest to identify unusual network behaviour, and Random Forest or XGBoost to classify known attack types. It also analyzes encrypted traffic using flow-based features and TLS fingerprints without decrypting user data. The proposed system is evaluated across multiple public intrusion detection datasets and real network traffic, with additional focus on detection accuracy, cross-dataset performance, system latency, resource usage, and automated attack blocking on constrained hardware.

Executive Summary

The increasing use of IoT devices has made small networks more exposed to cyber security threats. Most home and small-office routers provide basic network protection, but they lack advanced intrusion detection and behavioural analysis. This project proposes a Network Intrusion Detection and Prevention System (NIDPS) that combines multiple detection techniques to provide better protection against both known and previously unseen attacks.
The proposed system runs on a Linux-based device acting as the wireless access point of the protected network. All client traffic passes through this device before reaching the ISP gateway. The NIDPS follows a three-layer detection approach. Suricata is used for signature-based detection of known attacks, Isolation Forest is used to identify abnormal network behaviour, and Random Forest or XGBoost is used to classify known attack types. The outputs of these layers are combined to decide whether traffic should be passed, reported as suspicious, or blocked.
The system is designed to detect attacks such as SSH brute force, port scanning, command-and-control beaconing, ARP spoofing, DNS spoofing, data exfiltration, DDoS and lateral movement. Since modern network traffic is mostly encrypted, the system uses flow-based features and TLS fingerprints such as JA3 and JA3S instead of decrypting user traffic. This allows encrypted connections to be analysed while reducing privacy and performance concerns.
A major part of the project is the evaluation of machine learning models across different datasets. CIC-IDS-2017, CSE-CIC-IDS2018 and CIC-IoT-2023 are used for training and cross-dataset testing, while traffic collected from the actual deployment environment is used as an additional test set. The same flow extraction method is used across the datasets and the live system to reduce differences between training data and real network traffic.
When malicious activity is detected with sufficient confidence, the system can automatically block the responsible host using iptables and ipset. It also includes safeguards such as temporary blocks, allowlists and rate limits. A watchdog and configurable failure-handling mechanism are included so that the network remains in a defined state even if one of the detection components stops working.
The project is evaluated using both detection and system performance measures. These include precision, recall, F1-score, false positive rate, forwarding latency, throughput, detection latency, blocking time, resource usage and recovery behaviour. The overall aim is to develop a practical and affordable NIDPS that combines signature-based detection, machine learning, automated prevention and failure handling for small networks.

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Chapter 1

Introduction

Computer networks are used to connect devices, share data and access online services, but this connectivity also creates opportunities for malicious activity. A **network intrusion** occurs when an unauthorized user or program attempts to access, misuse, disrupt or damage a network or the devices connected to it. Intrusions may originate from outside the network, such as an attacker scanning for open ports or trying to gain access through weak credentials, or from inside the network through a compromised device or malicious user.

Network intrusions can take many forms. Some attacks focus on gaining unauthorized access, such as brute-force attacks, while others attempt to disrupt services through Denial-of-Service attacks. Attackers may also intercept or manipulate network communication through spoofing, move between devices after gaining access, transfer sensitive data outside the network, or communicate with compromised devices through command-and-control channels. Because these attacks behave differently, relying on a single detection method is often not enough.

Intrusion Detection and Prevention Systems are used to monitor network traffic and identify such malicious behaviour. Traditional systems commonly use predefined signatures or rules to recognize known attacks. These methods can be effective when an attack pattern is already known, but they may struggle with new attacks or variations that do not match an existing signature. Machine learning offers another approach by learning patterns in network traffic and identifying suspicious or abnormal behaviour. However, its effectiveness can depend heavily on the data used to train the model and the environment in which it is deployed.

Another challenge is that much of modern network communication is encrypted. While encryption protects user privacy, it also limits the amount of traffic content that can be directly inspected. This makes network security systems increasingly dependent on other information, such as traffic patterns, flow behaviour and connection metadata. These challenges create a need for intrusion detection systems that can combine multiple detection techniques and remain practical for real network environments.

The aim of this project is to develop a **Network Intrusion Detection and Prevention System (NIDPS)** for small networks. The project combines different detection approaches so that known attacks, abnormal behaviour and suspicious network activity can be identified and acted upon. The system is designed to operate on a Linux-based device placed directly in the network path, while also considering practical issues such as encrypted traffic, limited hardware resources and reliability.

The remainder of this report is organised as follows. **Chapter 2** presents the project scope, goals and objectives. **Chapter 3** reviews existing research and related work in network intrusion detection and machine learning. **Chapter 4** presents the Software Requirements

Specification (SRS) of the proposed system. **Chapter 5** explains the proposed approach and methodology used for detection, prevention and evaluation. **Chapter 6** describes the high-level and low-level design of the system.

1.1 Purpose of this Document

The purpose of this document is to present the design, development and evaluation of our Final Year Project, a **Network Intrusion Detection and Prevention System (NIDPS)** for small networks. The project aims to provide a practical security solution that can monitor network traffic, detect malicious or abnormal activity and respond to identified threats using a combination of signature-based detection and machine learning.

The main objective is to evaluate whether this hybrid approach can provide effective protection while remaining suitable for limited hardware. The research also examines how well machine learning models trained on public intrusion detection datasets perform when tested on different datasets and on real deployment traffic.

The main research questions focus on whether combining multiple detection techniques improves the detection of known and abnormal activity, whether the models can generalize across different network environments, and whether the complete system can operate efficiently without significantly affecting normal network performance.

This report covers the project scope, related research, software requirements, methodology and system design.

1.2 Intended Audience

This report is primarily intended for the **Final Year Project evaluation panel, project advisor, faculty members and other academic reviewers** who will assess the technical quality, design, implementation and evaluation of the project.

It may also be useful for **students, researchers and developers working in network security, intrusion detection or machine learning**, as it documents the proposed NIDPS architecture, detection methods, evaluation approach and practical deployment considerations.

The report is written with enough technical detail to explain the system clearly while keeping the discussion understandable for readers with a general background in computer science and cyber security.

1.3 Definitions, Acronyms, and Abbreviations

Table 1.1: Definitions, acronyms and abbreviations.

Term	Definition
NIDPS	Network Intrusion Detection and Prevention System
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
ML	Machine Learning
RF	Random Forest
XGBoost	Extreme Gradient Boosting
IoT	Internet of Things
C2	Command and Control
DoS	Denial of Service
DDoS	Distributed Denial of Service
TLS	Transport Layer Security
JA3	TLS client fingerprint generated from properties of the ClientHello message
JA3S	TLS server fingerprint generated from properties of the ServerHello message
ARP	Address Resolution Protocol
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
NAT	Network Address Translation
IP	Internet Protocol
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
ISP	Internet Service Provider
AP	Access Point
PCAP	Packet Capture file containing recorded network traffic
NFQUEUE	Linux Netfilter queue used to pass packets from the kernel to a user-space application for inspection
iptables	Linux firewall utility used to configure packet filtering and traffic-control rules
ipset	Linux utility used to maintain sets of IP addresses for efficient firewall matching and blocking
Suricata	Open-source network IDS/IPS used for signature-based traffic inspection
Isolation Forest	Unsupervised machine learning algorithm used to detect abnormal or unusual network traffic

Term	Definition
Flow	A group of related packets identified using network properties such as source and destination addresses, ports and protocol
Signature-Based Detection	Detection of attacks by matching network activity against predefined rules or known attack patterns
Anomaly Detection	Detection of activity that differs significantly from established normal network behaviour
Zero-Day Attack	An attack that exploits a previously unknown or unaddressed vulnerability or behaviour
API	Application Programming Interface
F1-Score	Evaluation metric that combines precision and recall into a single value
False Positive	Legitimate network activity that is incorrectly identified as malicious
Fail-Open	Failure mode in which network traffic is allowed to continue when a security component becomes unavailable
Fail-Closed	Failure mode in which network traffic is blocked when a security component becomes unavailable

1.4 Conclusion

The proposed NIDPS aims to improve the security of small networks by combining multiple intrusion detection techniques within a single system. The project focuses on detecting both known and abnormal network activity while also considering practical issues such as encrypted traffic, limited hardware resources and system reliability. Chapter 1 introduced the background, purpose and overall direction of the project. The following chapters discuss the related research, software requirements, proposed methodology and system design.

Chapter 2

Project Vision

This chapter explains the overall vision of the proposed Network Intrusion Detection and Prevention System. It describes what the system is intended to do, the problem it addresses, the goals of the project, its scope, major constraints, relevant stakeholders, and the broader value of the proposed solution. Since this project includes both research and development, it focuses on building a working NIDPS as well as evaluating how well the selected intrusion detection techniques perform in different network environments.

2.1 Problem Domain Overview

Our system implements a detection pipeline with three layers to tackle diverse attack vectors. The first layer uses Suricata with the Emerging Threats ruleset to capture known malicious activity via fixed signatures. The second layer uses an Isolation Forest model to flag baseline traffic anomalies that lack pre-existing rules. The third layer applies Random Forest or XGBoost classifiers to categorize flagged events into specific threat types. Instead of relying on a single indicator, the system aggregates findings across all three layers before taking action, either allowing the connection, logging an alert, or initiating a block.

To protect user privacy and minimize processing overhead, the system avoids decrypting packet payloads. Instead, it evaluates connection behavior through flow statistics and TLS parameters, specifically JA3 and JA3S fingerprinting. When malicious activity is confirmed, an automated response module updates local firewall configurations using iptables and ipset to block the origin host immediately. A web-based management interface provides administrators with system telemetry, security logs, and a list of currently blocked endpoints.

The research component focuses on model adaptability across changing network conditions. Machine learning models often suffer performance degradation when deployed outside their initial training data. To evaluate cross-environment resilience, we test our models on benchmark datasets as well as live traffic captured directly from our access point deployment.

2.2 Problem Statement

Existing intrusion detection approaches have their limitations. Signature based systems are effective against known attacks, but they may miss new attacks or modified versions of existing ones. Machine learning approaches can identify unusual traffic, but their performance

may drop when they are used on a network that is different from the dataset on which they were trained.
The problem addressed by this project is therefore the lack of an affordable and practical solution for small networks that can combine known-attack detection, anomaly detection, and attack classification while still running efficiently on limited hardware. The system must also be able to deal with encrypted traffic, avoid excessive false positives, and continue behaving safely if one of its components fails.

2.3 Problem Elaboration

The overall problem can be divided into smaller problems that need to be handled together.
First, no single detection method works for everything. Signature tools like Suricata catch known attacks instantly, but they miss brand new threats. On the flip side, machine learning spots weird behavior, but it can trigger false alarms on normal traffic. Using just one approach leaves major security gaps, so the system needs a multi-layer setup.
Second, machine learning models often struggle when moved to a new environment. A model might get perfect scores on one dataset, but completely fail on a different network with different devices and user habits. To test real-world reliability, we evaluate our models across multiple datasets, like CIC-IDS-2017, CSE-CIC-IDS2018, and CIC-IoT-2023, as well as live traffic from our own access point.
Third, most internet traffic is encrypted. Trying to decrypt network traffic slows down the hardware and invades user privacy. Instead of reading packet contents, the system looks at connection behavior, flow patterns, and unencrypted TLS handshake signatures like JA3 and JA3S.
Fourth, automatic blocking can backfire if the system makes a mistake. While blocking bad actors via iptables and ipset stops attacks fast, false alarms can accidentally lock out real users. To prevent this, the response engine includes safeguards like allowlists, temporary blocks, and adjustable threat thresholds.
Finally, the system runs on low power hardware. If the NIDPS uses too much CPU or memory, it will slow down internet speeds for everyone on the network. To keep things fast and stable, we use lightweight ML models, process data off-path, and set up fail-safes so a system crash doesn't knock out the entire internet connection.

2.4 Goals and Objectives

The main goal of this project is to build and evaluate a practical hybrid NIDPS that protects small networks using both signature matching and machine learning.
Key objectives include:

- **Hardware & Deployment:** Implement an inline NIDPS between client devices and the ISP gateway, complete with a watchdog and auto-recovery rules to prevent network downtime.

- **Multi-Layer Detection:** Combine Suricata (using Emerging Threats rules) for known attacks, Isolation Forest for unseen traffic anomalies, and Random Forest/XGBoost for classifying specific threats.
- **Encrypted Traffic Analysis:** Assess encrypted connections using flow statistics and TLS fingerprints (JA3/JA3S) without decrypting user payloads.
- **Automated Mitigation:** Automatically block confirmed threats using Linux firewall rules (iptables/ipset).
- **Cross-Environment Research:** Train and test models across CIC-IDS-2017, CSE-CIC-IDS2018, and CIC-IoT-2023, and benchmark their performance on live traffic from our actual access point.
- **Management & Telemetry:** Build a web dashboard for live monitoring, alerts, and system controls, measuring throughput, latency, CPU/memory usage, and detection accuracy under real traffic.

2.5 Project Scope

The system will operate directly as a wireless access point positioned between connected devices and the ISP gateway. Core features include:

- **Multi-Layer Detection:** Combines Suricata to catch known attack rules, Isolation Forest to spot unusual traffic spikes, and Random Forest or XGBoost to figure out the exact attack type.
- **Encrypted Traffic Analysis:** Looks at connection patterns and unencrypted TLS signatures to check if a stream is safe, completely skipping payload decryption to keep user privacy intact.
- **Active Response:** Automatically kicks bad actors off the network by blocking their IP, while giving admins a simple web dashboard to manage alerts and track blocked devices.
- **Research & Testing:** Benchmarks model accuracy across several benchmark datasets and real traffic captures from our setup, while making sure memory and CPU usage stay light on modest hardware.

Key Deliverables

- A working NIDPS access point prototype with automated prevention and fail-safe controls.
- The complete ML model training, testing, and cross-dataset evaluation pipeline.
- A web management dashboard.
- Practical performance benchmarks (latency, resource usage, detection accuracy).

- Full system design and research documentation.

Out of Scope

To maintain a clear focus, the project will not:

- Function as an enterprise-grade security suite or support enterprise-scale hosting.
- Monitor traffic from devices bypassing the access point.
- Decrypt user traffic payloads.
- Cover every existing exploit category or run continuous online model retraining on the live device.

2.6 Sustainable Development Goal (SDG)



Figure 2.1: UN Sustainable Development Goal targeted by the NIDPS.

Our project primarily aligns with the **Industry, Innovation and Infrastructure** SDG. Modern digital services rely heavily on secure and stable network connections. By adapting advanced threat detection for low-cost, resource-constrained hardware, this project offers an accessible way to make small scale networks more resilient against cyberattacks.

2.7 Constraints

Building and running this system comes with a few limitations:

- **Hardware Limits:** Low cost hardware cannot handle heavy software. Memory and CPU usage must stay low, which rules out heavy ML models.
- **Network Speed:** Sitting directly in the network path means any extra delay will slow down the internet. Traffic processing must be fast to keep connections smooth.
- **Encrypted Traffic:** Since packet contents are hidden, reading payloads is off the table. The ML side has to rely only on connection patterns and TLS details.

- **Data Limits:** Public security datasets do not cover every real-world scenario. That is why testing across multiple datasets and live traffic is necessary.
- **Coverage Gaps:** The system can only monitor devices that connect directly through its access point. Any device using a different network stays invisible.
- **Blocking Mistakes:** Blocking too aggressively locks out real users, but loose rules let threats through. Finding the right balance requires careful tuning and safety defaults.

2.8 Business Opportunity

Most small offices, research labs, and home users cannot afford expensive corporate security gear. They also lack dedicated IT teams to manage complicated setups. This opens up a clear opportunity for a simple, low cost security solution built on standard Linux hardware. By securing traffic right at the main access point, it protects every connected device at once, without needing extra software installed on each phone or laptop. While this project is centered on research and a working prototype, it builds a solid base for a practical product aimed at smaller networks.

2.9 Stakeholders Description/ User Characteristics

Network Owners and Admins: The primary users of the system which involve configuration of alerts, blocked devices, and system preferences. Since these are not security experts, the system is designed to provide informative feedback and employ safety defaults.
Regular Network Users: These are the individuals and devices which access the WI-FI network. They are not direct users of the dashboard but benefit from its existence as it helps protect and optimize the network.
The Project Team: This team is responsible for developing the system, including coding, training, and testing the algorithms.
Advisors and Evaluators: Faculty members who review the details of the research, evaluate the code, and ensure that the project meets its objectives.
Future Researchers: Potential students and developers who will review the project's details and possibly adopt it in their own research.

2.10 Conclusion

The contents of chapter 2 provided an overview of the proposed NIDPS project, which included the problem statement, major aspects and issues, and objectives of the intended solution. In this chapter, we have discussed the project scope, its key objectives, constraints, and stakeholders, as well as the potential benefits for business, and specified the relevant Sustainable Development Goal. Thus, this chapter sets the course and boundaries of the

project, providing the basis for the requirements, methods, design, and implementation that will be elaborated in subsequent chapters.

Chapter 3

Literature Review / Related Work

This chapter reviews recent research related to Network Intrusion Detection and Prevention Systems (NIDPS). It focuses on the detection techniques, datasets, machine learning models, deployment approaches and evaluation methods used in previous studies. Each study is summarised, critically analysed and related to the proposed system, and the chapter ends with summary tables of the reviewed literature, the datasets they use and the systems closest to the proposed NIDPS.

3.1 Definitions, Acronyms, and Abbreviations

The important definitions, acronyms and abbreviations used in this chapter are listed below:

- **AUC:** Area Under the (ROC) Curve
- **CIC:** Canadian Institute for Cybersecurity
- **DT:** Decision Tree
- **EIF:** Extended Isolation Forest
- **F1-score:** Harmonic mean of precision and recall
- **FPR:** False Positive Rate
- **IDS:** Intrusion Detection System
- **IIoT:** Industrial Internet of Things
- **IoT:** Internet of Things
- **iForest / IF:** Isolation Forest
- **IPS:** Intrusion Prevention System
- **LSTM:** Long Short-Term Memory (recurrent neural network)
- **ML:** Machine Learning
- **NIDPS:** Network Intrusion Detection and Prevention System
- **NIDS:** Network Intrusion Detection System
- **OC-SVM:** One-Class Support Vector Machine
- **RF:** Random Forest
- **SVM:** Support Vector Machine

- **XGBoost:** Extreme Gradient Boosting
- **Zeek:** Open-source network security monitor that produces connection logs

3.2 Detailed Literature Review

This section covers notable literature on hybrid signature and machine learning intrusion detection, unsupervised anomaly detection with Isolation Forest, supervised attack classification with tree ensembles, and the deployment and generalisation of ML-based detectors. Each item is presented with a summary, a critical analysis and its relationship to the proposed NIDPS.

3.2.1 Performance Comparison of Intrusion Detection Systems and Application of Machine Learning to Snort

Snort and Suricata are compared on a 10 Gbps network, and machine learning is added to Snort to reduce its false alarms. An SVM optimised with the firefly algorithm lowers the false positive rate to 8.6%.

3.2.1.1 Summary of the Research Item

Shah and Issac [1] compared the two most widely used open-source signature engines, Snort and Suricata, running on identical machines at a traffic rate of 10 Gbps. Suricata handled the high traffic speed with a lower packet drop rate because of its multi-threaded design, but used more computing resources, while Snort showed slightly higher detection accuracy. Since Snort produced a large number of false positive alarms, the authors developed an adaptive Snort plug-in and compared several learning algorithms to filter its alerts. SVM was selected, and the best configuration, an SVM optimised with the firefly algorithm, achieved a false positive rate of 8.6% and a false negative rate of 2.2%.

3.2.1.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It gives a practical, measured comparison of Snort and Suricata under heavy load, which is useful when choosing the signature engine for an inline system.
- It shows clearly that machine learning can reduce the false alarms of a signature engine.
- The SVM models were evaluated offline; SVMs are relatively heavy, and the work does not show the approach running on live traffic on low-cost hardware.
- The system only detects and does not block traffic, and it does not consider encrypted traffic.

3.2.1.3 Relationship to the Proposed Research Work

This work supports the use of Suricata as the first layer of the proposed NIDPS because it copes better with high traffic rates, and it supports the idea of using machine learning to reduce the false positives of a signature engine. Unlike this work, the proposed system uses lightweight tree models that classify flows in milliseconds and acts on the verdict by blocking the offending host.

3.2.2 A Novel Hybrid Intrusion Detection Method Integrating Anomaly Detection with Misuse Detection

A C4.5 decision tree (misuse detection) splits the normal training data into subsets, and a separate one-class SVM is trained on each subset for anomaly detection. Evaluated on NSL-KDD.

3.2.2.1 Summary of the Research Item

Kim, Lee and Kim [2] proposed a hierarchical hybrid intrusion detection method in which a misuse detection model and an anomaly detection model are combined in a decomposed structure. A C4.5 decision tree is first trained as the misuse detector, and it is also used to divide the normal training data into smaller, more uniform subsets. A one-class SVM is then trained on each subset, so every anomaly model describes one region of normal behaviour more precisely and indirectly benefits from the knowledge of known attacks. On the NSL-KDD dataset the method detected both known and unknown attacks better than conventional methods while keeping false positives low, and the anomaly model needed only about 50% of the training time and 60% of the testing time of the conventional models.

3.2.2.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- The decomposition of normal traffic gives more accurate anomaly models and noticeably lower training and testing time.
- It shows that relying on a single model to do all the work produces more false positives than letting each component handle the traffic it is best at.
- The evaluation uses only NSL-KDD, a dataset derived from 1998 traffic that does not represent modern or encrypted traffic.
- It is an offline detection method; no prevention or deployment on a real network is shown.

3.2.2.3 Relationship to the Proposed Research Work

The proposed NIDPS follows the same principle of combining misuse (signature) detection with anomaly detection so that each part handles the traffic it suits best. It replaces the

decision tree with Suricata rules and a Random Forest/XGBoost classifier, uses Isolation Forest instead of one-class SVM for cheaper anomaly scoring, and evaluates on modern datasets and live traffic.

3.2.3 Collaborative and Hybrid Network Intrusion Detection Framework Based on Suricata and Isolation Forest

Suricata detects known attacks and an Isolation Forest flags anomalies that may be unknown attacks. Several NIDS instances share alerts through a central log.

3.2.3.1 Summary of the Research Item

Chiba et al. [3] proposed a hybrid and collaborative NIDS framework. Suricata performs signature-based detection of known attacks and was chosen over Snort because it is multi-threaded and divides its workload across processing resources. Traffic that does not match any signature is examined by an Isolation Forest, which detects anomalies that may indicate unknown attacks. Since Isolation Forest does not depend on distance or density calculations, the authors argue that it keeps the computational cost low. The NIDS instances are collaborative: alerts are stored in a central log so that an attack detected by one sensor is also recognised by the others.

3.2.3.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- The pairing of Suricata with Isolation Forest is lightweight and matches the needs of edge devices.
- Sharing confirmed attacks between sensors lets the whole network benefit from one detection.
- It is mainly a design proposal: there is no attack-classification stage and only limited experimental evaluation.
- It does not describe automated blocking, failure handling or the analysis of encrypted traffic.

3.2.3.3 Relationship to the Proposed Research Work

This is one of the works closest to the proposed NIDPS, which also places Suricata first and sends traffic to an Isolation Forest. The proposed system extends this idea into a complete three-layer implementation by adding a supervised attack classifier, decision fusion, automated blocking and a watchdog, all running on low-cost hardware for home and small-office networks.

3.2.4 Self-Healing Hybrid Intrusion Detection System: An Ensemble Machine Learning Approach

A C5 decision tree acts as the signature stage, an LSTM detects anomalies, and features of detected anomalies are turned into new signatures that are fed back into the first stage.

3.2.4.1 Summary of the Research Item

Kushal et al. [4] proposed a three-stage self-healing hybrid IDS. In the first stage, a C5 decision tree trained on UNSW-NB15 classifies packets as normal or attack and acts as the signature-based detector. Packets that it does not match are passed to the second stage, an LSTM recurrent neural network trained on ADFA-LD, which flags deviations from normal behaviour. In the third stage, the attributes of anomalous packets are extracted and added to the training data of the C5 stage, so the system learns new signatures and can detect similar attacks earlier. The C5 classifier reached a true positive rate of about 97% with a false positive rate of about 8%, while the LSTM reached a detection rate of about 90% with a false alarm rate of 17% and an AUC of 0.936.

3.2.4.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- Turning detected anomalies into new signatures is an interesting way to keep the detector up to date without manual work.
- The two stages were evaluated separately on different datasets, and the authors note that the performance of the integrated system still needs to be assessed.
- The deep learning stage is computationally expensive and has a high false alarm rate, so wrongly learned signatures could classify dangerous traffic as normal or block benign traffic.
- Real-time testing on live networks is left as future work.

3.2.4.3 Relationship to the Proposed Research Work

The proposed NIDPS shares the goal of adapting to the local network, but keeps a fixed, verified set of models on the device and adapts only the anomaly threshold through calibration on normal traffic. This keeps the system adaptable while avoiding the cost and risk of learning new signatures online.

3.2.5 Comparative Evaluation of Machine Learning and Signature-Based NIDS for Multi-Class and Binary Threat Detection

Snort is compared with Random Forest, XGBoost and Decision Tree on UNSW-NB15. The ensemble models clearly outperform Snort in accuracy and false positive rate.

3.2.5.1 Summary of the Research Item

Wen-Udeoji, Muyeba and Mohammadi [5] built an empirical framework to compare the signature-based Snort NIDS with three machine learning models, Random Forest, XGBoost and Decision Tree, for both binary and multi-class threat detection on the UNSW-NB15 dataset while keeping its real class distribution. Random Forest reached 87% accuracy compared with 56.21% for Snort, with a false positive rate of 25% against 62.5% for Snort. Decision Tree and XGBoost reached 80.2% and 82.84% accuracy respectively. The authors conclude that the results support hybrid strategies that combine signature-based and machine learning detection.

3.2.5.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It is a recent, direct comparison of a signature engine with tree-based ML models on the same data.
- Keeping the natural class distribution gives a more realistic picture than artificially balanced data.
- Only one dataset is used, and the hybrid approach that the authors recommend is not implemented or evaluated.
- Real-time performance and resource use are not reported.

3.2.5.3 Relationship to the Proposed Research Work

The results support the choice of tree ensembles (Random Forest and XGBoost) for Layer 3 of the proposed NIDPS and the decision to keep the signature engine and the ML models as separate layers whose outputs are fused, rather than relying on either one alone.

3.2.6 Real-Time Hybrid Signature and Machine Learning Intrusion Detection

A signature engine is combined with deep learning models for real-time detection, and the decisions of both detection stages are combined after each stage has produced its own output.

3.2.6.1 Summary of the Research Item

Naik et al. [6] presented a real-time hybrid intrusion detection approach in which a signature-based engine detects known attacks and deep learning models detect anomalous behaviour. Each detection stage produces its own decision, and their experiments showed that combining these decisions at the end (late fusion) gives better results than passing the output of one stage as input to the other.

3.2.6.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It shows the benefit of late fusion of independent detector outputs in a real-time setting.
- Deep learning models need more processing power, which makes the approach harder to run inline on low-cost hardware.
- Automated prevention, encrypted traffic and cross-dataset evaluation are not addressed.

3.2.6.3 Relationship to the Proposed Research Work

The proposed NIDPS also uses late fusion across its three layers, but with optimised tree models instead of deep learning. Each layer's output is recorded, so the system can show which layer made the decision and why before deciding whether to pass, alert or block.

3.2.7 Improving Detection Accuracy of Network Intrusions Using a Hybrid NIDS Based on Isolation Forest and Random Forest

Suricata is combined with an Isolation Forest that flags anomalous traffic and a Random Forest that identifies the attack type. It reaches 96.33% accuracy on CIC-IDS-2017.

3.2.7.1 Summary of the Research Item

Wang and Avrianto [7] proposed a hybrid NIDS in which Suricata performs real-time packet inspection, an Isolation Forest flags anomalous traffic and a Random Forest identifies the family of the attack. The models were trained on pre-processed captured traffic and evaluated on CIC-IDS-2017 and on simulated attack data. The hybrid model reached 96.33% accuracy on CIC-IDS-2017 and 99.86% on the simulated data, and compared with standalone Suricata it detected more unknown threats with fewer false positives.

3.2.7.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It is the closest published design to the proposed system and shows that Isolation Forest and Random Forest work well together.
- The evaluation on CIC-IDS-2017 covers a limited subset of attacks, mainly DoS and port scanning.
- The very high accuracy is reported on simulated data, which may not reflect real network traffic.
- The system detects but does not block traffic, and failure handling is not considered.

3.2.7.3 Relationship to the Proposed Research Work

This is the most similar approach to the proposed NIDPS. The proposed system keeps the Isolation Forest and Random Forest pairing, but places Suricata inline in IPS mode as the first layer, adds automated blocking and failure handling, and is evaluated on three public datasets together with a self-collected dataset.

3.2.8 Extended Isolation Forest for Intrusion Detection in Zeek Data

The hyperparameters of the Extended Isolation Forest are tuned to detect malicious connections in Zeek logs. The simplest configuration (extension level 0) performs best.

3.2.8.1 Summary of the Research Item

Moomtaheen et al. [8] tuned the hyperparameters of the Extended Isolation Forest (EIF), a variant of Isolation Forest, to detect malicious traffic in Zeek connection logs. The authors created a new dataset, UWF-ZeekDataFall22, and used Hadoop for storage and Apache Spark for training. The model reached accuracies of 82.3% for the Resource Development tactic, 82.21% for Reconnaissance and 78.3% for Discovery. The best results came from extension level 0, which is equivalent to the standard Isolation Forest.

3.2.8.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It uses real Zeek connection logs labelled with MITRE ATT&CK tactics, which is closer to real deployments than older benchmarks.
- It shows that the extended variant gives no benefit over the basic Isolation Forest on this data.
- Accuracy around 80% is moderate, so Isolation Forest alone is not enough for reliable detection.
- The big-data pipeline (Hadoop and Spark) is not suited to small, resource-constrained devices.

3.2.8.3 Relationship to the Proposed Research Work

This study supports the decision to use the standard Isolation Forest in Layer 2 of the proposed NIDPS rather than a more complex variant, and to combine it with a signature layer and a supervised classifier because its accuracy on its own is moderate.

3.2.9 An Optimized Isolation Forest Based Intrusion Detection System for Heterogeneous and Streaming Data in IIoT Networks

An Isolation Forest tuned with a modified Harris Hawks Optimisation (ERHHO) reduces features and training time and handles streaming data with concept drift.

3.2.9.1 Summary of the Research Item

Elsaid and Binbusayyis [9] proposed OIFIDS, an Isolation Forest based intrusion detection system for Industrial IoT networks. The Isolation Forest is optimised with ERHHO, a modified Harris Hawks Optimisation algorithm, which reduces the number of features and the training time. The system was evaluated on NSL-KDD, UNSW-NB15 and CSE-CIC-IDS2018, reaching accuracies of 95.6%, 94.8% and 99% respectively. On streaming data with concept drift it achieved AUC values of 0.948, 0.922 and 0.97 on the three datasets.

3.2.9.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It addresses streaming traffic and concept drift, which are real problems for a deployed detector.
- Feature reduction lowers computational complexity, which suits resource-constrained devices.
- The optimisation step adds complexity to training, and the results are still on public benchmarks rather than a live deployment.

3.2.9.3 Relationship to the Proposed Research Work

The proposed NIDPS also has to cope with traffic that changes over time. Instead of a metaheuristic optimiser, it handles this with feature selection and a periodic recalibration of the Isolation Forest threshold on the local network's normal traffic.

3.2.10 Improved Isolation Forest with Sparse Auto-Encoder Feature Extraction

A sparse auto-encoder extracts features and a new split-point rule improves the Isolation Forest, giving better accuracy, recall, F1-score and AUC on KDDCUP99.

3.2.10.1 Summary of the Research Item

Yu et al. [10] improved the Isolation Forest for intrusion detection in two ways. First, a sparse auto-encoder is used to extract better features from the raw traffic records. Second, a new rule is used to choose the split points of the isolation trees. Evaluated on the KDD-CUP99 dataset, the improved model achieved higher accuracy, recall, F1-score and AUC than the standard Isolation Forest.

3.2.10.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It shows that feature quality has a large effect on how well Isolation Forest separates attacks.
- KDDCUP99 is outdated and contains many redundant records, so the results may not hold on modern traffic.
- The auto-encoder adds training and inference cost compared with the plain Isolation Forest.

3.2.10.3 Relationship to the Proposed Research Work

This work confirms that the proposed NIDPS should invest in good flow features for its Isolation Forest. The proposed system obtains them from a consistent flow extractor and feature selection rather than from a neural auto-encoder, keeping the anomaly layer lightweight.

3.2.11 An Improved X-Means and Isolation Forest Based Methodology for Network Traffic Anomaly Detection

X-iForest removes the need to know the anomaly ratio in advance by clustering the suspected anomalies. It improves average AUC by 8.1% over the basic Isolation Forest.

3.2.11.1 Summary of the Research Item

Feng et al. [11] addressed a common problem of Isolation Forest: the contamination parameter (the share of anomalies) must be guessed, although the true ratio is unknown in real traffic. Their method, X-iForest, first runs an Isolation Forest with a deliberately high contamination value, then measures the distance of each suspected anomaly from the centre of the normal data, removes extreme values and uses X-means and 2-means clustering to separate real anomalies from normal points. Across four simulated and four public datasets, the average AUC was 8.1% higher than the basic Isolation Forest, and the method had the best AUC on six of the eight datasets.

3.2.11.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It removes the need to guess the contamination parameter, which is a real practical issue.
- The evaluation datasets are general anomaly-detection datasets and simulated Gaussian data, not network intrusion datasets.
- Performance is weaker on small datasets, and the method has not been tested on live traffic.

3.2.11.3 Relationship to the Proposed Research Work

The proposed NIDPS faces the same problem of setting the anomaly threshold without knowing the true anomaly ratio. It solves it by calibrating the Isolation Forest threshold on a period of normal traffic from the local network, which is simpler to run on the device.

3.2.12 Intrusion Detection: A Comparison Study of Machine Learning Models Using Unbalanced Dataset

Five classifiers are compared on the imbalanced UNSW-NB15 dataset. Random Forest and XGBoost give the best overall results.

3.2.12.1 Summary of the Research Item

Ajagbe, Awotunde and Florez [12] compared five machine learning models, Linear SVC, Logistic Regression, Random Forest, Decision Tree and XGBoost, on the UNSW-NB15 dataset, in which about 92.7% of the records are normal and 7.3% are attacks. Random Forest and XGBoost performed best in accuracy, precision, recall and F1-score. The AUC-ROC was 0.91 for Random Forest and Linear SVC, 0.90 for XGBoost, 0.83 for Decision Tree and 0.70 for Logistic Regression.

3.2.12.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It evaluates the models on imbalanced data, which reflects real networks where attacks are rare.
- The way oversampling and undersampling were applied is not clearly described.
- The study was not tested in a real-world setting and uses a single dataset.

3.2.12.3 Relationship to the Proposed Research Work

This comparison supports the choice of Random Forest and XGBoost for the attack classifier (Layer 3) of the proposed NIDPS, and highlights class imbalance as an issue that the training pipeline must handle.

3.2.13 Enhanced Intrusion Detection Systems Performance with UNSW-NB15 Data Analysis

Four supervised classifiers are compared after correlation-based feature selection. Random Forest performs best with an F1-score of 97.80%.

3.2.13.1 Summary of the Research Item

More et al. [13] compared Logistic Regression, SVM, Decision Tree and Random Forest on UNSW-NB15 after exploratory data analysis, correlation-based feature selection and random sampling. Random Forest was the best model, with an F1-score of 97.80%, an accuracy of 98.63% and a false alarm rate of 1.36%.

3.2.13.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It shows strong Random Forest results with a low false alarm rate after simple feature selection.
- Results come from a single dataset, so they say little about generalisation to other networks.
- Detection latency and resource use are not reported.

3.2.13.3 Relationship to the Proposed Research Work

The study supports the use of Random Forest in Layer 3 of the proposed NIDPS. The proposed system goes further by also measuring latency and resource use and by testing the model across datasets.

3.2.14 An Optimized Gradient Boosting Framework for IoT Intrusion Detection on the CICIoT2023 Dataset

XGBoost, LightGBM and CatBoost are evaluated on CICIoT2023 with a focus on edge deployment. Reducing 46 features to 23 cuts inference latency by 30–51% with almost no loss of accuracy.

3.2.14.1 Summary of the Research Item

Almahaqeri et al. [14] proposed an optimised gradient boosting intrusion detection framework for IoT environments, with an emphasis on practical deployment at the edge. An intrusion detection system on an edge gateway must not only detect attacks accurately but also run efficiently with limited computing power. The framework was evaluated on the CICIoT2023 dataset, comparing XGBoost, LightGBM and CatBoost for binary, attack-family (8-class) and fine-grained (34-class) classification.
Training is performed offline, while the deployed edge system performs online flow classification. To reduce the processing cost, the authors apply gain-based feature selection and reduce the 46 features to 23, which lowered inference latency by around 30–51% and training time by 33–40% with almost no loss in predictive performance. XGBoost achieved 99.61% accuracy and a macro F1-score of 0.995 in binary detection, taking 0.355 microseconds per sample.

3.2.14.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It considers deployment efficiency in addition to detection accuracy, which many intrusion detection works ignore.
- It shows that reducing the number of features greatly improves inference speed without sacrificing accuracy, which suits edge environments.
- The evaluation is limited to CICIoT2023, so generalisation to other networks is not shown; the authors list cross-dataset validation and streaming conditions as future work.
- Edge suitability is judged from latency measurements rather than a full deployment on a low-power gateway, and only flow-level features are used.

3.2.14.3 Relationship to the Proposed Research Work

This work is closely related to the proposed NIDPS, which also performs detection on a resource-constrained Linux gateway. Like the paper, the proposed system separates offline training from online inference and treats latency and resource use as evaluation criteria. Unlike this study, it also includes signature-based and anomaly detection, cross-dataset evaluation, automated blocking, failure handling and testing on traffic from the actual deployment.

3.2.15 Cross-Dataset Generalisation of Deep Learning Intrusion Detection Models

Deep IDS models that score about 99% on the dataset they were trained on drop to roughly 50% when tested on a different dataset.

3.2.15.1 Summary of the Research Item

Xin and Xu [15] examined how well intrusion detection models generalise beyond the dataset they were trained on. They report that deep IDS models which reach around 99% on their own dataset often fall to about 50%, close to random guessing, when evaluated on other datasets. This shows that high benchmark scores do not guarantee good performance on a different network.

3.2.15.2 Critical Analysis of the Research Item

The strengths and weaknesses of this work are:

- It exposes a major weakness of single-dataset evaluation, which most reviewed studies rely on.
- It focuses on detection accuracy and does not propose a deployable system or measure runtime cost.

3.2.15.3 Relationship to the Proposed Research Work

This finding is the main reason why the proposed NIDPS is trained on CIC-IDS-2017, CSE-CIC-IDS2018 and CIC-IoT-2023 with a common flow extractor, evaluated across datasets, and validated on traffic collected from the deployment network itself.

3.3 Literature Review Summary Table

Table 3.1 gives a general overview of the studies reviewed in Section 3.2, listing the method, dataset, main results and limitations of each work.

Table 3.1: Summary of the reviewed literature.

Title	Year	Method	Dataset	Results	Limitations
Performance comparison of IDSs and application of ML to Snort [1]	2018	Snort vs Suricata at 10 Gbps; SVM (firefly-optimised) filtering Snort alerts	Test-bed traffic (10 Gbps)	FPR 8.6%, FNR 2.2%; Suricata handles high load better	Offline SVM; no blocking
Hybrid IDS integrating anomaly with misuse detection [2]	2014	C4.5 decision tree + multiple one-class SVMs	NSL-KDD	Higher detection of known and unknown attacks; ~50% of training time	Old dataset; offline only

Title	Year	Method	Dataset	Results	Limitations
Collaborative hybrid NIDS based on Suricata and Isolation Forest [3]	2019	Suricata + Isolation Forest; alerts shared via central log	Not reported (design proposal)	Lower computational cost (design)	No classifier; limited evaluation
Self-healing hybrid IDS: an ensemble ML approach [4]	2024	C5 decision tree + LSTM; anomalies turned into new signatures	UNSW-NB15, ADFA-LD	C5: ~97% TPR, 8% FPR; LSTM: 90% DR, AUC 0.936	Stages tested separately; costly DL
Comparative evaluation of ML and signature-based NIDS [5]	2025	Snort vs RF, XGBoost, DT	UNSW-NB15	RF 87% vs Snort 56.21% accuracy; FPR 25% vs 62.5%	Single dataset; hybrid not built
Real-time hybrid signature and ML detection [6]	2025	Signature engine + deep learning, late fusion	Not reported	Late fusion better than chaining stages	Heavy DL models; no prevention
Hybrid NIDS based on Isolation Forest and Random Forest [7]	2025	Suricata + Isolation Forest + Random Forest	CIC-IDS-2017, simulated	96.33% (CIC-IDS-2017), 99.86% (simulated)	Small attack subset; no blocking
Extended Isolation Forest for intrusion detection in Zeek data [8]	2024	Extended Isolation Forest, Spark	UWF-Zeek-DataFall22	78.3–82.3% accuracy; level 0 (basic IF) best	Moderate accuracy; big-data stack
Optimized Isolation Forest IDS for IIoT networks [9]	2024	Isolation Forest tuned with ERHHO; streaming data	NSL-KDD, UNSW-NB15, CSE-CIC-IDS2018	Accuracy 95.6%, 94.8%, 99%; streaming AUC up to 0.97	Complex optimisation; no live test
Improved Isolation Forest based on split points [10]	2024	Sparse auto-encoder features + new split rule	KDDCUP99	Higher accuracy, recall, F1 and AUC than IF	Outdated dataset; extra cost
X-means and Isolation Forest for traffic anomaly detection [11]	2022	X-iForest: IF + distance + X-means clustering	4 simulated + 4 public sets	Average AUC +8.1% over IF; best on 6 of 8	Not intrusion data; weak on small sets
Comparison of ML models using unbalanced dataset [12]	2024	Linear SVC, LR, RF, DT, XGBoost	UNSW-NB15	RF and XGBoost best; AUC 0.91 (RF), 0.90 (XGB)	Single dataset; not tested in real setting

Title	Year	Method	Dataset	Results	Limitations
Enhanced IDS performance with UNSW-NB15 data analysis [13]	2024	LR, SVM, DT, RF with correlation feature selection	UNSW-NB15	RF: F1 97.80%, accuracy 98.63%, FAR 1.36%	Single dataset; no latency
Optimized gradient boosting framework on CICIoT2023 [14]	2026	XGBoost, LightGBM, CatBoost; gain-based feature selection	CICIoT2023	XGBoost 99.61%, macro F1 0.995; 30–51% lower latency	One dataset; no device deployment
Cross-dataset generalisation of deep IDS models [15]	2025	Transformer IDS with calibration; cross-dataset evaluation	NSL-KDD, UNSW-NB15, CIC-IDS2017	~99% within-dataset drops to near chance across datasets	No deployable system

3.4 Data Set Summary Table

Table 3.2 gives an overview of the datasets used in the reviewed studies and how relevant each one is to the proposed NIDPS.

Table 3.2: Datasets used in the reviewed studies.

Sr. No	Dataset	Year	Environment / Traffic	Used in	Relevance to the proposed NIDPS
1	KDDCUP99	1999	Simulated military network (DARPA 1998)	[10]	Historical baseline only; outdated and redundant
2	NSL-KDD	2009	Cleaned version of KDDCUP99	[2, 9]	Still based on 1998 traffic; no modern attacks
3	UNSW-NB15	2015	Hybrid real and synthetic traffic, 9 attack families	[4, 5, 9, 12, 13]	Common benchmark; class imbalance
4	ADFA-LD	2013	Host-based Linux system-call traces	[4]	Host-level data; not used for flows
5	CIC-IDS-2017	2017	5 days of enterprise-style traffic	[7]	Used for training and cross-evaluation
6	CSE-CIC-IDS2018	2018	Larger AWS-hosted network, 10 days	[9]	Used for training and cross-evaluation
7	CICIoT2023	2023	105 IoT devices in a smart-home topology	[14]	Closest to home/IoT target; used for training
8	UWF-Zeek-DataFall22	2022	Zeek connection logs labelled with MITRE ATT&CK	[8]	Shows value of real Zeek-style logs

3.5 Comparison with Closest Existing Systems

The systems reviewed above that come closest to the proposed NIDPS are compared in Table 3.3. The table shows which detection layers each system uses and whether it addresses inline blocking, encrypted traffic, evaluation on multiple datasets or real traffic, low-cost hardware and failure handling.

Table 3.3: Comparison of the closest existing systems with the proposed NIDPS.

System	Signature layer	Anomaly layer	Classifier	Inline blocking	Encrypted traffic	Multi-dataset / real traffic	Low-cost hardware	Failure handling
Wang and Avrianto [7]	Yes (Suricata)	Yes (IF)	Yes (RF)	No	No	No	No	No
Chiba et al. [3]	Yes (Suricata)	Yes (IF)	No	Proposed	No	Not evaluated	No	No
Kim, Lee and Kim [2]	Misuse model (C4.5)	Yes (OC-SVM)	Yes (DT)	No	No	No (NSL-KDD)	No	No
Kushal et al. [4]	Learned (C5)	Yes (LSTM)	Yes (C5)	No	No	Partial	No	No
Wen-Udeoji et al. [5]	Yes (Snort)	No	Yes (RF, XGB, DT)	No	No	No	No	No
Naik et al. [6]	Yes	Yes (deep)	n/r	n/r	No	n/r	No	No
Shah and Issac [1]	Yes (Snort)	No	Yes (SVM)	No	No	n/r	No (server)	No
Proposed NIDPS	Yes (Suricata IPS)	Yes (IF, calibrated)	Yes (RF/XGB)	Yes (ipset)	Yes (JA3, flow stats)	Yes (3 public + own)	Yes (Raspberry Pi 4)	Yes (watchdog, 3 modes)

n/r: not reported. IF: Isolation Forest, RF: Random Forest, XGB: XGBoost, DT: Decision Tree, OC-SVM: One-Class SVM.

The existing systems mainly address one or two of these issues, whereas the proposed system improves on them in five ways:

- **Complete system:** Previous works either propose a design or only detect traffic. The proposed system detects, blocks and applies safeguards, while keeping forwarding latency close to that of a regular router.
- **Three layers with fusion:** Suricata catches known attacks first, an Isolation Forest tuned to local traffic spots anomalous behaviour, and a Random Forest/XGBoost classifier identifies the attack family. Their outputs are fused so that the system records what each layer decided and why.
- **Lightweight models:** Unlike many deep learning systems, the proposed system uses lightweight tree models on the forwarding path, and latency, throughput and resource use are measured.
- **Cross-dataset evaluation:** Because most models fail on unseen datasets, the system is trained on CIC-IDS-2017, CSE-CIC-IDS2018 and CIC-IoT-2023 using one flow extractor and validated on self-collected household traffic.

- **Encrypted traffic and failure handling:** The system analyses encrypted connections using JA3/JA3S fingerprints and flow metadata without decryption, and uses a watchdog with configurable fail-open, fail-closed and degraded modes. None of the reviewed systems addresses either problem.

The contribution of the proposed project is therefore not the idea of combining signature detection with machine learning, which has been researched extensively, but turning that idea into a practical, deployable system for home and small-office networks.

3.6 Conclusion

The reviewed literature shows a clear direction: signature engines such as Snort and Suricata detect known attacks reliably but miss new ones and raise many false alarms, while machine learning detects unknown behaviour but is rarely deployed inline. Hybrid systems that let each technique handle the traffic it suits best, and that fuse their decisions at the end, give the best results [2, 5, 6]. Isolation Forest is a cheap and effective anomaly detector when it is given good features and a sensible threshold [8, 10, 11], and tree ensembles such as Random Forest and XGBoost are the strongest and fastest choice for classifying known attack types [12, 13, 14].
The review also exposed common gaps. Most studies use a single, often outdated dataset, and models that score close to 99% on one dataset can fall to about 50% on another [15]. Few works report latency or resource use, few implement automated blocking, and none of the reviewed systems analyses encrypted traffic or defines what happens when a component fails. The proposed NIDPS is designed around these gaps: it combines Suricata, a calibrated Isolation Forest and a Random Forest/XGBoost classifier with late fusion, blocks confirmed attackers automatically, analyses encrypted traffic through flow metadata and JA3/JA3S fingerprints, is evaluated across three public datasets and real traffic, and runs on low-cost hardware with defined failure modes.

Chapter 4

Software Requirement Specifications

This chapter specifies the requirements of the proposed Network Intrusion Detection and Prevention System (NIDPS). It describes what the system must do, how well it must do it, and the conditions under which it operates, in enough detail for the system to be built and tested against it. The chapter begins with the list of features and the functional requirements, grouped by module. It then states the quality attributes and the measurable non-functional requirements, followed by the assumptions on which the specification rests. The main interactions between users and the system are described as use cases. The chapter closes with the hardware and software requirements, the design of the graphical user interface, the database design and an analysis of project risks. Requirements are given identifiers (for example FR-2.1) so that they can be traced to the design in Chapter 6 and to the tests in Section 6.6.6.
The system has five classes of users or actors, which are referred to throughout this chapter:

- **Network Administrator:** the owner of the home or small-office network who installs the NIDPS and manages it through the web dashboard. This user is not assumed to be a security expert.
- **Network User (client device):** any phone, laptop or IoT device that connects to the NIDPS access point. These users do not interact with the NIDPS directly; they are protected by it.
- **Attacker:** an external host or a compromised device inside the network that generates malicious traffic. The attacker triggers detection and prevention use cases.
- **System Scheduler / Watchdog:** internal time- and event-driven actors that run scheduled jobs (rule updates, threshold calibration, data purging) and supervise the health of every component.
- **Model Maintainer:** a project team member who trains and validates machine learning models offline and publishes model packages to the device.

4.1 List of Features

The NIDPS provides the following features. Each feature is refined into functional requirements in Section 4.2.

Table 4.1: Features of the proposed NIDPS.

ID	Feature	Description
NIDPS-1	Inline wireless access point	Acts as the visible network in the system. All devices connect to it. It provides WiFi, DHCP, DNS and NAT before passing the connection on to the router.
NIDPS-2	Signature-based detection (Layer 1)	Checks every packet that passes through using Suricata
NIDPS-3	Flow extraction and TLS fingerprinting	Groups up related packets and assigns them 78 features. It can identify details from encrypted packets as well.
NIDPS-4	Anomaly detection (Layer 2)	An Isolation Forest will be trained on normal traffic relative to the local network it is placed in and will be able to detect an anomaly score for traffic.
NIDPS-5	Attack classification (Layer 3)	Once attacks or anomalies have been detected, they are then reused and classified into specific families through Random Forest or XGBoost.
NIDPS-6	Decision fusion	Collects the decisions from each layer and decides whether traffic may pass through or the device should be blocked due to frequent anomalous traffic.
NIDPS-7	Automated prevention	It will block identified hosts that are dangerous by blocking their IP address, there will be a specific whitelist as well for unique cases. Features eventual expiry as well.
NIDPS-8	Fault tolerance	Takes care of all components with a watchdog system, if some system fails the watchdog may close the network, or let traffic pass through free depending on configuration.
NIDPS-9	Monitoring	An interface available on the network's LAN which will show statistics and alerts including all configuration settings.
NIDPS-10	Configurations	The administrator can edit or turn off the layers, change thresholds for detection, change failsafe actions and manage black/whitelists.
NIDPS-11	Logs and storage	Stores alerts, traffic logs, configurations in files and databases.
NIDPS-12	Training and versioning	The system can train and test its models using the collected data. Each version is tagged so it can be returned to.

ID	Feature	Description
NIDPS-13	Auto maintenance	Rules and tracking features are updated daily and thresholds updated. Older data outside of the limit deleted.

4.2 Functional Requirements

The functional requirements below describe what the system must do. They are grouped by the features listed in Section 4.1, and each requirement is numbered according to its feature.

4.2.1 NIDPS-1: Inline Wireless Access Point

FR-1.1 The system shall act as a wireless access point with a configurable network name (SSID) and WPA2/WPA3 password.

FR-1.2 The system shall assign IP addresses to connected devices using DHCP.

FR-1.3 The system shall act as the only DNS server for connected devices.

FR-1.4 The system shall pass client traffic on to the ISP router using NAT over its wired connection.

FR-1.5 The system shall stop wireless clients from talking to each other directly, so that traffic between them also passes through the NIDPS.

4.2.2 NIDPS-2: Signature-Based Detection (Layer 1)

FR-2.1 The system shall send all forwarded traffic to Suricata so it can be checked before it is passed on.

FR-2.2 The system shall check every packet against the Emerging Threats Open ruleset.

FR-2.3 The system shall drop packets that match rules set to "drop" and allow all other packets through.

FR-2.4 The system shall use custom rules to detect SSH brute force, port scans, DNS spoofing, SYN floods and known malicious TLS fingerprints.

FR-2.5 The system shall detect ARP spoofing by comparing ARP replies with the devices it has assigned addresses to.

FR-2.6 The system shall pass every alert, with its source, destination, rule and severity, on to the other layers.

4.2.3 NIDPS-3: Flow Extraction and TLS Fingerprinting

FR-3.1 The system shall group related packets into flows, where each flow is one conversation between two devices in both directions.

FR-3.2 The system shall end a flow when the connection closes or after a set period of inactivity (default 30 seconds) or activity (default 10 seconds).

FR-3.3 The system shall calculate 78 features for each flow as packets arrive, without storing the packets themselves.

FR-3.4 The system shall read identifying details (JA3, JA3S and SNI) from encrypted TLS connections and attach them to the flow, without decrypting the traffic.

4.2.4 NIDPS-4: Anomaly Detection (Layer 2)

FR-4.1 The system shall check that each flow matches the features expected by the current model and reject any that do not.

FR-4.2 The system shall give each flow an anomaly score between 0 and 1 using an Isolation Forest.

FR-4.3 The system shall mark a flow as anomalous when its score is above the anomaly threshold.

FR-4.4 The system shall set the anomaly threshold based on normal traffic from the local network, leaving out traffic linked to alerts or blocked devices.

FR-4.5 The system shall learn how much data each device normally sends out and flag it when a device sends far more than usual (possible data theft).

FR-4.6 The system shall flag devices that contact the same destination at regular intervals with similar-sized messages (possible C2 beaconing).

4.2.5 NIDPS-5: Attack Classification (Layer 3)

FR-5.1 The system shall sort each flow into one of the following classes using Random Forest or XGBoost: Benign, Reconnaissance, Brute Force, DoS/DDoS, Botnet/C2, Spoofing, or Infiltration/Lateral Movement.

FR-5.2 The system shall give a confidence value for each classification.

FR-5.3 The system shall label a flow as "unknown" when the model's confidence is too low, instead of forcing it into a class.

4.2.6 NIDPS-6: Decision Fusion

FR-6.1 The system shall combine the results of all three layers to decide whether each flow should be passed (PASS), flagged (ALERT) or blocked (BLOCK).

FR-6.2 The system shall record which layers led to each decision and the evidence used.

FR-6.3 The system shall keep a risk score for each device that rises with suspicious activity and drops over time when the device behaves normally.

FR-6.4 The system shall block a device when its risk score reaches the block threshold.

4.2.7 NIDPS-7: Automated Prevention

FR-7.1 The system shall block a dangerous device by adding its IP address to a block list.

FR-7.2 The system shall remove each block automatically once its time runs out.

FR-7.3 The system shall never block devices on the allowlist, which includes the NIDPS itself, the router, the DNS server and the administrator's devices.

FR-7.4 The system shall double the block time each time the same device is blocked again within 24 hours, up to a set maximum.

FR-7.5 The system shall stop blocking and only raise alerts if too many blocks happen within one minute, and warn the administrator.

FR-7.6 The system shall log every block, skipped block and release.

4.2.8 NIDPS-8: Fault Tolerance

FR-8.1 The system shall have a watchdog that checks each component every second.

FR-8.2 The system shall treat a component as failed if it misses three checks in a row or stops running, and restart it automatically.

FR-8.3 The system shall, while a component is down, either let all traffic through (fail-open), block all traffic (fail-closed) or run with reduced checking (degraded), depending on configuration.

FR-8.4 The system shall switch to degraded mode and check only some flows when it becomes overloaded, and return to normal once the load drops.

FR-8.5 The system shall log when each failure happened, which mode was used, when it recovered and how many packets were affected.

4.2.9 NIDPS-9: Monitoring

FR-9.1 The system shall require the administrator to log in before the dashboard can be viewed.

FR-9.2 The system shall show live traffic statistics, alerts, blocks and types of attacks, updated at least every two seconds.

FR-9.3 The system shall show a list of alerts with time, source, destination, attack type, layer and confidence, which can be filtered.

FR-9.4 The system shall show the evidence behind each alert.

FR-9.5 The system shall list blocked devices with the reason and time left, and let the administrator unblock them.

FR-9.6 The system shall show traffic and risk scores for each connected device.

FR-9.7 The system shall show the health of each component, along with CPU, memory, temperature and uptime.

4.2.10 NIDPS-10: Configurations

FR-10.1 The system shall let the administrator turn each detection layer on or off while running.

FR-10.2 The system shall let the administrator choose the failure mode and watchdog settings.

FR-10.3 The system shall let the administrator change detection thresholds, block times and flow timeouts within safe limits.

FR-10.4 The system shall let the administrator add and remove devices from the allowlist.

FR-10.5 The system shall let the administrator reload the rules and recalibrate the anomaly threshold on demand.

FR-10.6 The system shall apply configuration changes without restarting, and log who made each change and what was changed.

4.2.11 NIDPS-11: Logs and Storage

FR-11.1 The system shall store alerts, decisions, blocks, risk scores, calibration data, health events and configuration in a local database.

FR-11.2 The system shall not store packet contents during normal operation.

FR-11.3 The system shall let the administrator export alerts and blocks for a chosen time period as a CSV file.

4.2.12 NIDPS-12: Training and Versioning

FR-12.1 The system shall build training data using the same flow extractor that runs on the device.

FR-12.2 The system shall train and test models offline.

FR-12.3 The system shall save each model as a package that includes the model, its feature list, extractor version, classes, training date and test results.

FR-12.4 The system shall refuse to load a model that does not match the current flow extractor.

FR-12.5 The system shall keep the previous model so the administrator can return to it in one step.

4.2.13 NIDPS-13: Auto Maintenance

FR-13.1 The system shall download and test new rules every day, and keep using the old rules if the test fails.

FR-13.2 The system shall recalibrate the anomaly threshold every week.

FR-13.3 The system shall delete flow records older than the retention period (default 7 days).

4.3 Quality Attributes

The quality attributes below describe the general qualities the NIDPS must have and the design decisions that provide them. The measurable targets for each attribute are given as non-functional requirements in Section 4.4.

Table 4.2: Quality attributes of the NIDPS.

Attribute	Meaning for the NIDPS	How it is achieved
Performance	The NIDPS must not noticeably slow the network and must react to attacks quickly.	Only the kernel and Suricata sit in the forwarding path; ML runs on flows beside the path; lightweight tree models; ipset lookups in constant time.
Availability	A failure of the NIDPS must not leave the household without internet unless the owner chose that.	Fail-open default enforced by the kernel; automatic service restarts; blocks that expire on their own.
Reliability	Detection must keep working under load and recover from faults in a defined way.	Watchdog with heartbeats; degraded mode with hysteresis; per-service isolation; recovery events logged.
Security	The NIDPS itself must not become an easy target or a tool against its own users.	Authenticated LAN-only dashboard; allowlist; block rate limit; models loaded only from a protected directory; no pickle loading for XGBoost.
Privacy	Household traffic must stay private.	No TLS decryption; no payload storage; only metadata and alerts kept, with a retention limit.
Accuracy	Attacks must be detected with few false alarms.	Three complementary layers; decision fusion requiring strong or corroborated evidence; per-network threshold calibration.

Attribute	Meaning for the NIDPS	How it is achieved
Usability	A non-expert must be able to understand alerts and act on them.	Plain-language alert explanations; safe defaults; one-click unblock; clear health indicators.
Maintainability	Rules, models and thresholds must be easy to update.	Versioned model packages with rollback; separate custom rule file; single configuration file; modular services.
Extensibility	New detectors and responses must be addable without rewriting the system.	Common Detector interface; publish/subscribe event bus; JA3 extraction isolated so it can be replaced by JA4.
Testability	Each layer and the full pipeline must be measurable.	Layers can be toggled at runtime for ablation; PCAP replay; fault injection; every decision logged with its evidence.
Portability	The system should run on different low-cost Linux devices.	Standard Debian-based Linux components; Python services; no hardware-specific code apart from the wireless driver.

4.4 Non-Functional Requirements

The non-functional requirements set measurable targets for how well the system must perform. The numbers apply to the reference setup: a Raspberry Pi 4 with 4 GB RAM serving a home network of up to 30 devices. These are design targets that will be tested, and adjusted if needed, during the evaluation in Section 5.10.

4.4.1 Performance

- **NFR-1** The system shall add no more than 200 ms of delay (median round-trip time) under normal household use, with all layers turned on.
- **NFR-2** The system shall handle at least 50 Mbps of traffic with all layers turned on, and the maximum speed it can handle shall be measured and reported.
- **NFR-3** The system shall score each flow with the machine learning layers (Layers 2 and 3) in no more than 10 ms (median).
- **NFR-4** The system shall apply a block within 1 second of deciding to block a device.
- **NFR-5** The system shall produce a machine learning decision for long-running connections no later than 1 second after the active timeout.

4.4.2 Capacity

- **NFR-6** The system shall support at least 30 connected devices and at least 5 people using the dashboard at the same time.

4.4.3 Resource Use

- **NFR-7** The system shall use less than 70% CPU and less than 3 GB of memory on average under normal household use, leaving room for traffic spikes.

4.4.4 Reliability and Availability

- **NFR-8** The system shall detect a failed component within 3 seconds and restart it within 30 seconds in at least 95% of tested failures.
- **NFR-9** The system shall keep the internet connection working without interruption in fail-open mode when any detection component fails.

4.4.5 Accuracy

- **NFR-10** The system shall raise alerts on no more than 1% of normal traffic after calibration, and shall not block any normal device during a one-week test.
- **NFR-11** The system shall detect attacks more accurately (higher macro F1-score) than Suricata on its own, using the self-collected test data.

4.4.6 Security

- **NFR-12** The system shall only allow dashboard access from the local network, require a login, store passwords securely as salted hashes, and lock an account for 5 minutes after 5 wrong password attempts.
- **NFR-13** The system shall run all detection services without admin privileges, except those that need network access to work.

4.4.7 Privacy

- **NFR-14** The system shall not save packet contents or decrypted data during normal operation, and shall delete flow records after the retention period (default 7 days).

4.4.8 Usability

- **NFR-15** The system shall allow a non-expert user to find out why a device was blocked and unblock it within 1 minute, without needing a manual.
- **NFR-16** The system shall include a plain-language description of the suspected attack and the layer that detected it with every alert.

4.4.9 Maintainability

- **NFR-17** The system shall allow new models and rules to be installed, or rolled back, without interrupting network traffic.

4.4.10 Extensibility and Reusability

- **NFR-18** The system shall allow a new detector to be added without changing any other component.
- **NFR-19** The system shall allow the flow extractor and training pipeline to be reused separately from the device.

4.4.11 Portability

- **NFR-20** The system shall install with a single script on any 64-bit Debian-based ARM or x86 device that meets the hardware requirements in Section 4.7.1.

4.5 Assumptions

The specification is based on the following assumptions. If one of them does not hold, the affected requirements may need to be revisited.

- The NIDPS is the only wireless network that client devices use. The ISP router's own Wi-Fi is disabled, and devices connected by cable directly to the ISP router are outside the protection of the system.
- The host has at least a quad-core 64-bit CPU, 4 GB of RAM, a wireless adapter that supports access point mode and an Ethernet port for the uplink.
- The ISP router works correctly and provides internet access, but it is not relied on for any security function.
- The NIDPS host and the administrator's devices are trusted and have not been compromised.
- The administrator is not a security expert. Default settings must therefore be safe without tuning.
- Periodic internet access is available for ruleset updates.
- Most traffic is encrypted with TLS or QUIC, so payload inspection is not possible for the ML layers. Detection of encrypted traffic relies on flow metadata and handshake fingerprints only.
- The public datasets (CIC-IDS-2017, CSE-CIC-IDS2018 and CIC-IoT-2023) and their raw PCAP files remain available for download.

- Attack traffic for the self-collected dataset is generated only on an isolated network owned by the project team, against the team's own devices.
- Traffic observed during the calibration window is mostly benign; flows flagged by Suricata during this window are excluded.
- The household traffic volume does not exceed the measured capacity of the device for long periods; short peaks are handled by degraded mode.

4.6 Use Cases

This section describes the main use cases of the NIDPS. Figure 4.1 shows the use case diagram with the actors introduced at the beginning of this chapter. The use cases that represent the central functionality of the system, namely automatic detection and blocking, failure handling and the administrator's main tasks on the dashboard, are then described in detail in tabular form. In each table, the Basic Flow is written as a dialogue between the actor and the system, and the Alternative Flow describes exceptions; step numbers with a suffix (for example 4-A) replace the step of the basic flow with the same number.

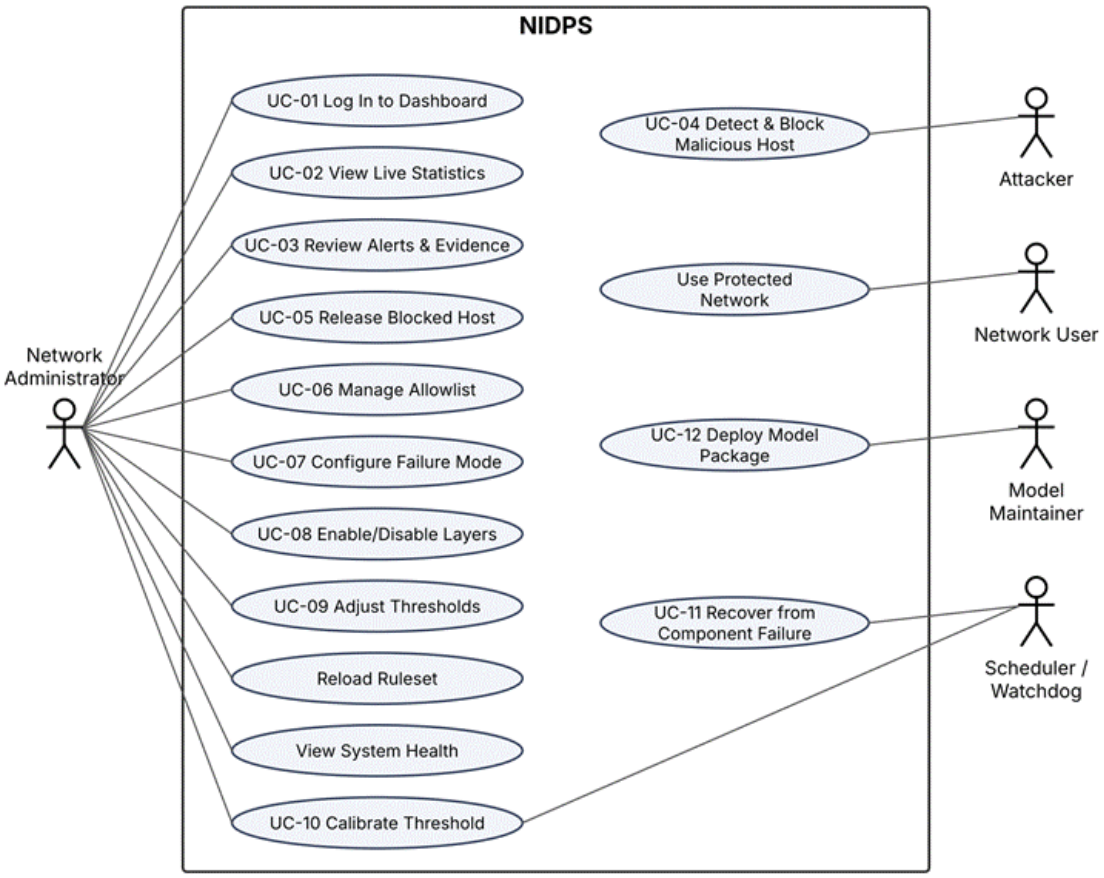


Figure 4.1: Use case diagram of the NIDPS.

Table 4.3: Use case UC-01: Log In to Dashboard.

Name		**UC-01: Log In to Dashboard**	
Actors		Network Administrator	
Summary		The administrator enters a username and password on the login page. After successful verification, the system establishes a session and shows the overview page.	
Pre-Conditions		The administrator's device is connected to the NIDPS network (LAN side). An administrator account exists. The administrator is not already logged in.	
Post-Conditions		An authenticated session is established and the overview page is displayed.	
Special Requirements		Passwords are stored as salted hashes. The account is locked for 5 minutes after 5 consecutive failed attempts (NFR-12).	
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens the dashboard address in a browser.	2	The system displays the login page asking for username and password.
3	The administrator enters a valid username and password and presses Log In.	4	The system verifies the credentials, creates a session, records the login in the audit log and displays the overview page.
Alternative Flow			
3	The administrator enters an invalid username or password.	4-A	The system shows the message "Incorrect username or password" and increments the failed-attempt counter.
3	The administrator enters wrong credentials for the fifth consecutive time.	4-B	The system locks the account for 5 minutes and shows the time remaining.

Table 4.4: Use case UC-02: View Live Network Statistics.

Name	**UC-02: View Live Network Statistics**
Actors	Network Administrator
Summary	The administrator views real-time throughput, flows per second, alert and block counts per detection layer, attack category distribution and per-device traffic.

Pre-Conditions		The administrator is logged in (UC-01).	
Post-Conditions		Live statistics are displayed and keep updating while the page is open.	
Special Requirements		Statistics refresh at least every 2 seconds (FR-9.2).	
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens the Overview page.	2	The system displays current throughput, flows per second, alerts and blocks per layer and the attack category chart for the last hour.
3	The administrator selects a different time window (for example 24 hours).	4	The system reloads the charts and counters for the selected window.
5	The administrator selects a client device from the device list.	6	The system shows that device's traffic summary and current risk score.
Alternative Flow			
2	No traffic has been observed in the selected window.	2-A	The system shows empty charts with the message "No traffic in this period".
	The live connection to the server is lost.		The system marks the data as stale and retries the connection automatically.

Table 4.5: Use case UC-03: Review Alerts and Evidence.

Name		**UC-03: Review Alerts and Evidence**	
Actors		Network Administrator	
Summary		The administrator browses the alert list, filters it and opens an alert to see which layer detected it, the evidence behind the verdict and the originating flow.	
Pre-Conditions		The administrator is logged in (UC-01).	
Post-Conditions		The selected alert and its evidence are displayed. The alert may be marked as reviewed.	
Special Requirements		Each alert includes a plain-language description of the suspected attack (NFR-16).	
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens the Alerts page.	2	The system displays the most recent alerts with time, source, destination, attack class, detecting layer, confidence and verdict.

3	The administrator filters by severity, layer, host or time range.	4	The system displays only the matching alerts.
5	The administrator selects an alert.	6	The system shows the evidence panel: Suricata signature, anomaly score and threshold, class probabilities, host risk history and the flow record.
7	The administrator marks the alert as reviewed.	8	The system records the review in the audit log.
Alternative Flow			
2	No alerts match the filter.	2-A	The system shows "No alerts match the selected filters".
3	The flow record has been deleted by the retention policy.	3-A	The system shows the stored evidence and notes that the flow details are no longer available.

Table 4.6: Use case UC-04: Detect and Block Malicious Host.

Name	**UC-04: Detect and Block Malicious Host**		
Actors	Attacker (initiator), System		
Summary	Traffic from an attacker is detected by one or more layers. The decision engine issues a BLOCK verdict, and the enforcer adds the attacker's address to the block set for a limited time and notifies the dashboard.		
Pre-Conditions	The NIDPS is running with at least one detection layer enabled. The attacker's traffic passes through the NIDPS.		
Post-Conditions	The attacker's address is in the block set with an expiry time. An audit record and a dashboard event exist for the block.		
Special Requirements	Blocking must take effect within 1 second of the verdict (NFR-4). Allowlisted addresses are never blocked (FR-7.3).		
Basic Flow			
Actor Action		**System Response**	
1	The attacker sends malicious traffic (for example an SSH brute-force attempt or a beaconing connection).	2	Suricata inspects the packets inline; the flow extractor builds flow records with features and TLS fingerprints.

	—	3	Layer 1 raises an alert and/or Layers 2 and 3 score the flow; the evidence is published on the event bus.
	—	4	The decision engine applies the verdict algorithm and returns BLOCK with the responsible layer(s).
	—	5	The enforcer checks the allowlist and rate limit, computes the block duration, adds the address to the ipset and writes an audit record.
6	The attacker continues sending traffic.	7	Packets from the attacker are dropped in the kernel before reaching Suricata. The dashboard shows the new block.
Alternative Flow			
3	Evidence is weak (for example an anomaly score just above τ only).	3-A	The verdict is ALERT; evidence is added to the host's risk score, and the host is blocked only if its score reaches Θ.
4	The attacker's address is on the allowlist.	4-A	The block is suppressed and logged as "block suppressed"; an alert is still shown.
4	The block rate limit R_max has been reached.	4-B	The verdict is downgraded to ALERT and a "block rate limit" warning is shown on the dashboard.
4	The attacker was blocked before in the last 24 hours.	4-C	The block duration is doubled for each earlier offence, up to the maximum.

Table 4.7: Use case UC-05: Release Blocked Host.

Name	**UC-05: Release Blocked Host**
Actors	Network Administrator
Summary	The administrator reviews the list of blocked hosts and manually releases a host that was blocked by mistake.
Pre-Conditions	The administrator is logged in (UC-01). At least one host is currently blocked.
Post-Conditions	The host's address is removed from the block set and the release is recorded in the audit log.
Special Requirements	The released host may optionally be added to the allowlist (UC-06).
Basic Flow	

Actor Action		System Response	
1	The administrator opens the Blocked Hosts page.	2	The system lists blocked hosts with address, device name, attack class, detecting layer, block start and remaining time.
3	The administrator selects a host and presses Release.	4	The system asks for confirmation and offers to add the host to the allowlist.
5	The administrator confirms.	6	The system removes the address from the ipset, marks the block as released, writes an audit record and refreshes the list.
Alternative Flow			
3	The block expired while the dialog was open.	3-A	The system informs the administrator that the block has already expired.
3	The administrator cancels.	3-B	No change is made.

Table 4.8: Use case UC-06: Manage Allowlist.

Name	**UC-06: Manage Allowlist**		
Actors	Network Administrator		
Summary	The administrator adds or removes IP or MAC addresses that must never be blocked, such as the gateway or the administrator's own laptop.		
Pre-Conditions	The administrator is logged in (UC-01).		
Post-Conditions	The allowlist is updated and immediately used by the enforcer.		
Special Requirements	The NIDPS host, gateway and DNS resolver are permanent entries and cannot be removed.		
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens the Allowlist section and presses Add.	2	The system shows a form for address and description.
3	The administrator enters an address and description and saves.	4	The system validates the address, stores the entry, publishes the change on the bus and records it in the audit log.
Alternative Flow			
2	The address is invalid or already listed.	2-A	The system shows a validation error and does not save.
2	The address is currently blocked.	2-B	The system asks whether to release the block as well (UC-05).

Table 4.9: Use case UC-07: Configure Failure Mode.

Name		**UC-07: Configure Failure Mode**	
Actors		Network Administrator	
Summary		The administrator chooses how the network behaves when a detection component fails: fail-open (keep internet access), fail-closed (stop forwarding until detection is restored) or degraded operation under load.	
Pre-Conditions		The administrator is logged in (UC-01).	
Post-Conditions		The new failure mode is saved and applied to the kernel queue configuration.	
Special Requirements		Changing between fail-open and fail-closed briefly reloads the inspection queue rule; forwarding is not interrupted.	
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens Settings and selects the Failure Mode tab.	2	The system shows the current mode with a description of its trade-off.
3	The administrator selects a new mode and presses Save.	4	The system asks for confirmation, explaining the effect of the chosen mode.
5	The administrator confirms.	6	The system updates the queue-bypass setting, saves the configuration, logs the change and shows a success message.
Alternative Flow			
3	The system cannot apply the new rule.	3-A	The previous mode remains active and an error message is shown.

Table 4.10: Use case UC-08: Enable or Disable Detection Layers.

Name	**UC-08: Enable or Disable Detection Layers**
Actors	Network Administrator
Summary	The administrator turns individual detection layers (Suricata, Isolation Forest, classifier) on or off. This is also used to run the layer ablation experiments.
Pre-Conditions	The administrator is logged in (UC-01).

Post-Conditions		The selected layers are active or inactive and the dashboard shows their status.	
Special Requirements		At least one layer must remain enabled unless the administrator explicitly confirms monitoring-only operation.	
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens Settings and selects the Detection Layers tab.	2	The system shows each layer with its status and a toggle.
3	The administrator switches a layer off and saves.	4	The system publishes the change; the decision engine stops using that layer's evidence and the dashboard marks it as disabled.
5	The administrator switches the layer back on.	6	The system resumes using the layer's evidence.
Alternative Flow			
2	The administrator tries to disable all layers.	2-A	The system warns that the network will be unprotected and requires explicit confirmation.

Table 4.11: Use case UC-09: Adjust Detection Thresholds.

Name		**UC-09: Adjust Detection Thresholds**	
Actors		Network Administrator	
Summary		The administrator changes detection sensitivity, such as the calibration percentile, classifier confidence thresholds, host block threshold and block durations.	
Pre-Conditions		The administrator is logged in (UC-01).	
Post-Conditions		The new thresholds are validated, saved and used by running services.	
Special Requirements		Each value has a validated range and a documented default, which can be restored with one action.	
Basic Flow			
Actor Action		**System Response**	
1	The administrator opens Settings and selects the Thresholds tab.	2	The system shows current values, defaults and allowed ranges.

3	The administrator changes a value and presses Save.	4	The system validates the value, saves it, publishes the change and logs old and new values.
Alternative Flow			
2	The value is outside the allowed range.	2-A	The system rejects the value and shows the allowed range.
2	The administrator presses Restore Defaults.	2-B	The system restores all default values after confirmation.

Table 4.12: Use case UC-10: Calibrate Anomaly Threshold.

Name	**UC-10: Calibrate Anomaly Threshold**		
Actors	System Scheduler, Network Administrator		
Summary	The anomaly threshold τ is recalculated from the scores of clean local traffic so that it fits the deployment network. It runs weekly or on demand.		
Pre-Conditions	The Isolation Forest model is loaded. Flow records from the calibration window are available.		
Post-Conditions	A new calibration state (τ, timestamp, sample count) is stored and used by Layer 2.		
Special Requirements	Flows linked to Suricata alerts or blocked hosts are excluded, and τ cannot move more than δ from the global threshold (Algorithm 5.4).		
Basic Flow			
Actor Action		**System Response**	
1	The scheduler triggers calibration (or the administrator presses Recalibrate Now).	2	The system collects anomaly scores of the clean flows from the calibration window.
	—	3	The system computes the p-th percentile, clips it to the allowed range, stores the new τ and publishes it on the bus.
	—	4	The dashboard shows the new τ, its date and the number of flows used.
Alternative Flow			
2	Fewer than N_min clean flows are available.	2-A	The system keeps the global threshold and reports that more traffic is needed.

Table 4.13: Use case UC-11: Recover from Component Failure.

Name		**UC-11: Recover from Component Failure**	
Actors		Watchdog	
Summary		When a component crashes, freezes or is overloaded, the watchdog detects it, applies the configured failure mode, restarts the component and records the recovery.	
Pre-Conditions		The watchdog is running and receiving heartbeats.	
Post-Conditions		The component is running again, or the system remains in the configured failure mode and the administrator is notified.	
Special Requirements		Detection within 3 seconds and restart within 30 seconds (NFR-8). Fail-open is enforced by the kernel even if the watchdog itself fails.	
Basic Flow			
Actor Action		**System Response**	
	—	1	A component misses three consecutive heartbeats or its process exits.
	—	2	The watchdog declares it failed, applies the failure mode and logs the failure event.
	—	3	systemd restarts the component; the watchdog waits for healthy heartbeats.
	—	4	The watchdog marks the component as running, restores normal mode and records the recovery time and packets dropped or bypassed.
5	The administrator views System Health.	6	The system shows the failure event and recovery details.
Alternative Flow			
3	The component fails to restart after the maximum number of attempts.	3-A	The system stays in the failure mode and shows a critical warning on the dashboard.
1	Load exceeds the high watermark without a crash.	2-A	The watchdog enters degraded mode and returns to normal when load falls below the low watermark.

Table 4.14: Use case UC-12: Deploy Model Package.

Name	**UC-12: Deploy Model Package**
Actors	Model Maintainer

Summary		The maintainer trains and validates a new model offline and deploys it to the device as a versioned package. The device checks compatibility before using it.	
Pre-Conditions		A model package has been produced by the offline training pipeline (FR-12.3). The maintainer has deployment access to the device.	
Post-Conditions		The new model is active and the previous package is kept for rollback.	
Special Requirements		XGBoost models are stored in native JSON format; packages are loaded only from the protected registry directory.	
Basic Flow			
Actor Action		**System Response**	
1	The maintainer copies the package to the model registry and runs the deploy command.	2	The system checks the feature list and extractor version against the running extractor.
	—	3	The system loads the new model, keeps the previous one, and shows the new version and its metrics on the dashboard.
Alternative Flow			
1	The feature schema does not match.	2-A	The system refuses the package, keeps the current model and reports the mismatch.
2	The maintainer runs the rollback command.	2-B	The system reactivates the previous package.

4.7 Hardware and Software Requirements

The NIDPS has two environments: the deployment environment, which is the device that runs inline on the network, and the development environment, which is used to build the software, train the models and generate test attacks. The requirements of both are listed below.

4.7.1 Hardware Requirements

Table 4.15: Hardware requirements.

Component	**Minimum / reference specification**	**Purpose**
NIDPS host	Raspberry Pi 4 Model B, 4 GB RAM (reference); any quad-core 64-bit ARM or x86 device with $\geq$ 4 GB RAM	Runs the access point, Suricata, flow extractor, ML scorer, decision engine, enforcer, watchdog and dashboard
Storage	32 GB microSD (A2 class) or SSD	Operating system, rulesets, model packages, event database and logs
Wireless adapter	Built-in Wi-Fi or USB Wi-Fi adapter supporting AP mode (802.11ac recommended)	Client-facing access point (wlan0)
Wired uplink	Gigabit Ethernet port and Cat5e/Cat6 cable	Connection to the ISP router (eth0)
Cooling and power	Official 5 V/3 A power supply; heatsink or fan case	Stable operation under sustained load without thermal throttling
ISP router / ONT	Existing ISP device (e.g. Huawei HG8546M), built-in Wi-Fi disabled	Upstream internet gateway
Attack host	Laptop or VM running Kali Linux	Generating controlled attacks on the isolated test network
Test client devices	At least two phones/laptops and one IoT device	Benign traffic, victims for lateral movement and spoofing tests
Training workstation	Laptop/PC with $\geq$ 16 GB RAM (32 GB recommended), $\geq$ 200 GB free disk	Regenerating datasets from PCAPs and training models offline

4.7.2 Software Requirements

Table 4.16: Software requirements.

Category	**Software (version)**	**Purpose**
Operating system	Raspberry Pi OS 64-bit or another Debian-based Linux with systemd	Base platform for the NIDPS host
Access point	hostapd, dnsmasq	Wi-Fi access point, DHCP and DNS
Firewall and queueing	iptables (nftables backend), ipset, NFQUEUE kernel support	Routing, NAT, inline queue, blocking and failure modes

Category	Software (version)	Purpose
Signature IDS/IPS	Suricata 7 or later; Emerging Threats Open ruleset; suricata-update	Layer 1 inline detection, JA3/JA3S logging
Programming language	Python 3.11 or later	All custom services and training scripts
ML libraries	scikit-learn (Isolation Forest, Random Forest), XGBoost, NumPy, pandas, imbalanced-learn, joblib	Model training and inference
Packet processing	Scapy / libpcap bindings; CICFlowMeter-compatible feature definitions	Flow extraction and TLS handshake parsing
Event bus and storage	Redis (Streams); SQLite 3	Messaging between services; event database
Dashboard	FastAPI (or Flask) with WebSocket support; HTML/CSS/JavaScript with a charting library	Web interface and API
Supervision	systemd units and hardware watchdog	Process supervision and automatic restart
Testing and evaluation	pytest, tcpreplay, iperf3, Wireshark/tcpdump, Kali Linux tools (nmap, hydra, hping3, arpspoof, dnsspoof)	Unit, integration, performance and live attack tests
Development tools	Git and GitHub, VS Code, Jupyter Notebook	Version control, coding and data analysis
Client side	Any modern web browser (Chrome, Firefox, Edge, Safari)	Accessing the dashboard

4.8 Graphical User Interface

The NIDPS has one graphical user interface: the web dashboard used by the Network Administrator. It is served by the NIDPS host and is reachable only from the protected network. Network users have no interface, because protection is transparent to them. The interface has six main screens.

4.8.1 Login Screen

The login screen authenticates the administrator before any information is shown. Error messages do not reveal whether the username or the password was wrong, and the account

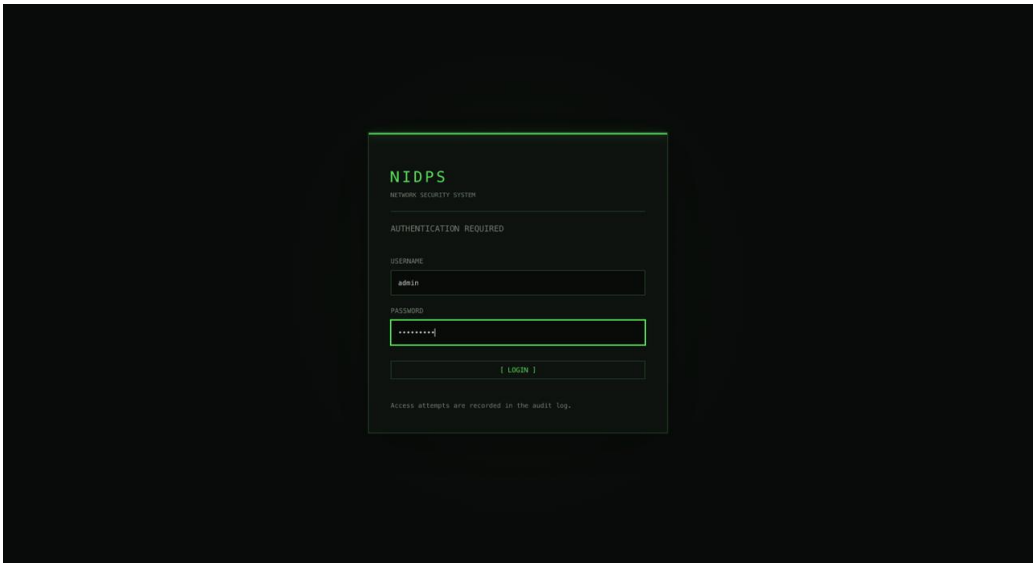


Figure 4.2: Login screen.

is locked after repeated failures. Covers: UC-01.

4.8.2 Overview Screen

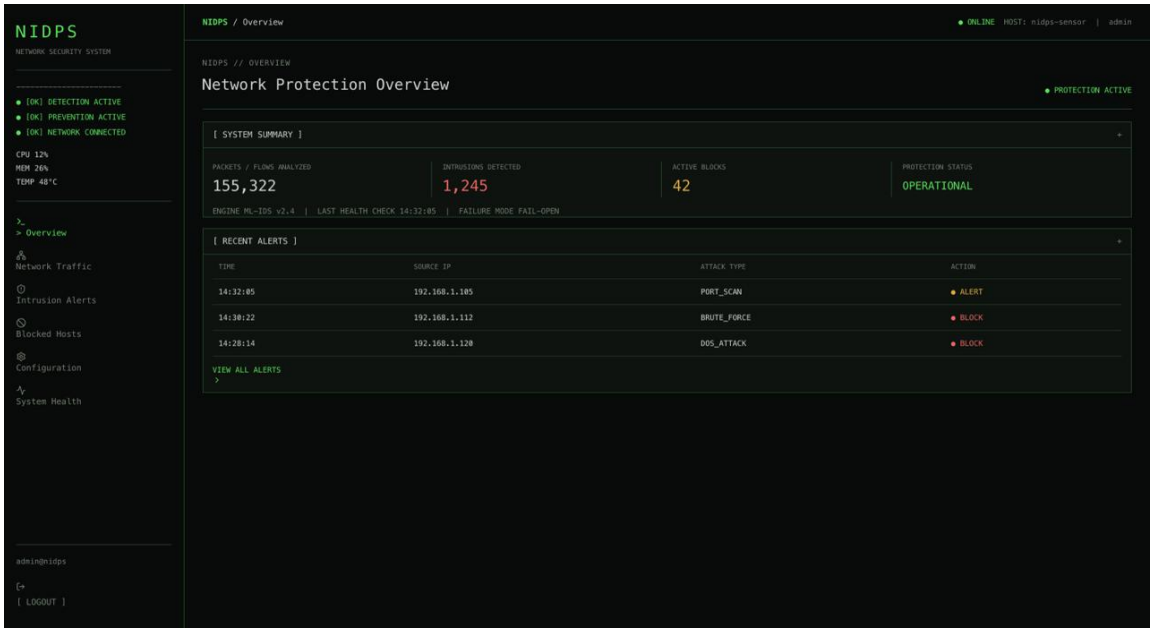


Figure 4.3: Overview screen.

The overview screen is the home page. It answers the questions "is my network safe right now?" and "is the NIDPS working?" at a glance. Counters and charts update live. Selecting a device shows its traffic summary and risk history. Covers: UC-02.

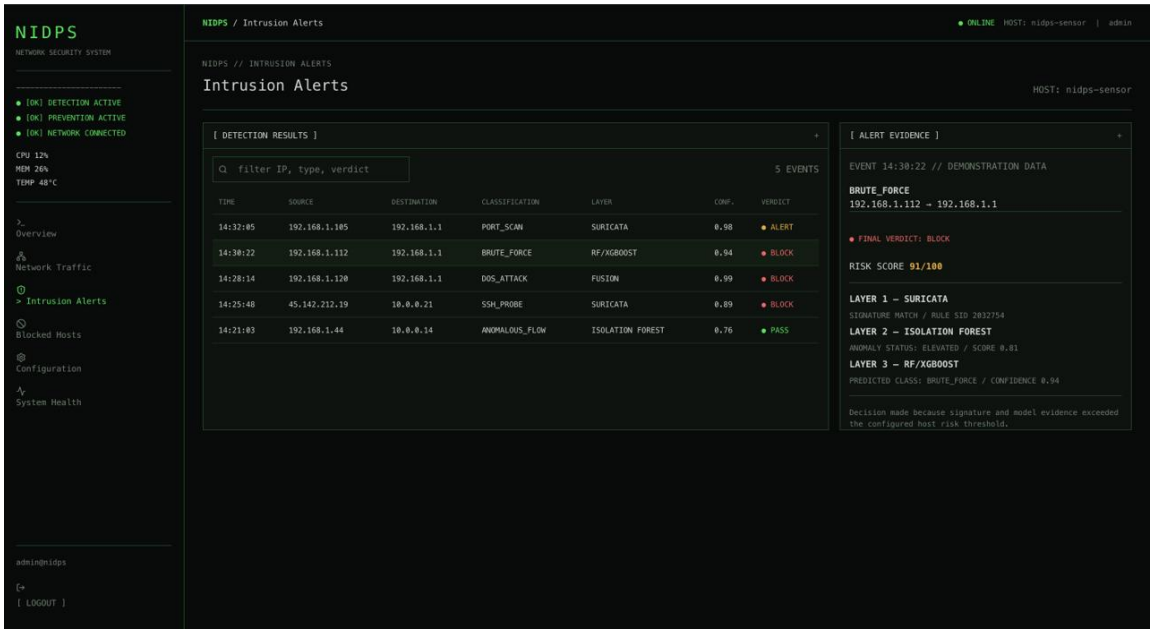


Figure 4.4: Alerts screen with the evidence panel.

4.8.3 Alerts Screen

The alerts screen lists every ALERT and BLOCK verdict. Selecting an alert opens the evidence panel, which explains in plain language what was detected and shows the evidence from each layer, so the administrator can judge whether the alert is genuine. Covers: UC-03, UC-04 (results).

4.8.4 Blocked Hosts and Allowlist Screen

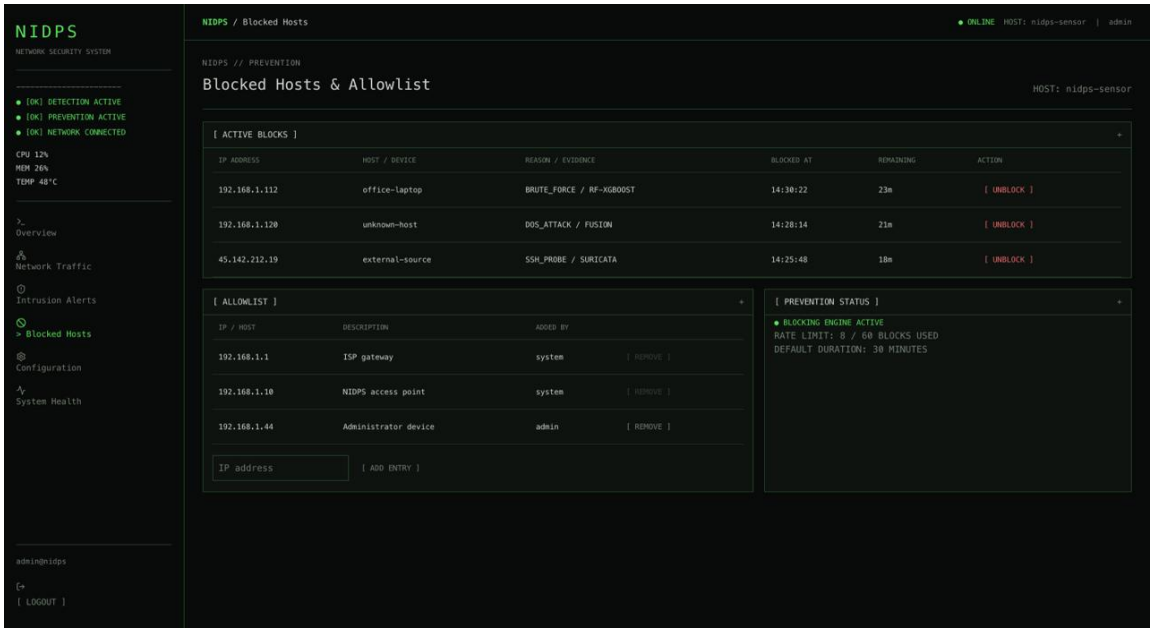


Figure 4.5: Blocked Hosts and Allowlist screen.

This screen lets the administrator see who is blocked and why, release hosts blocked by mistake and maintain the allowlist. A warning banner appears here if the block rate limit has been reached. Covers: UC-05, UC-06.

4.8.5 Settings Screen

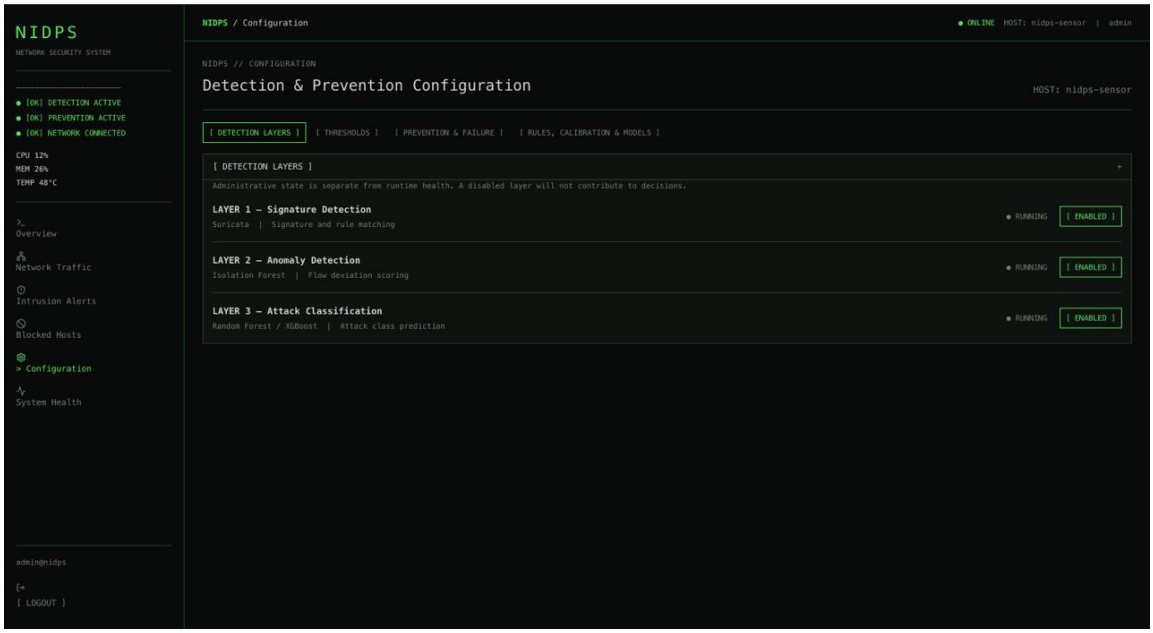


Figure 4.6: Settings screen.

The settings screen groups all runtime configuration. Every change is validated, applied without restarting services and recorded in the audit log. Current values are shown next to their defaults. Covers: UC-07, UC-08, UC-09, UC-10, UC-12 (model version display).

4.8.6 System Health Screen

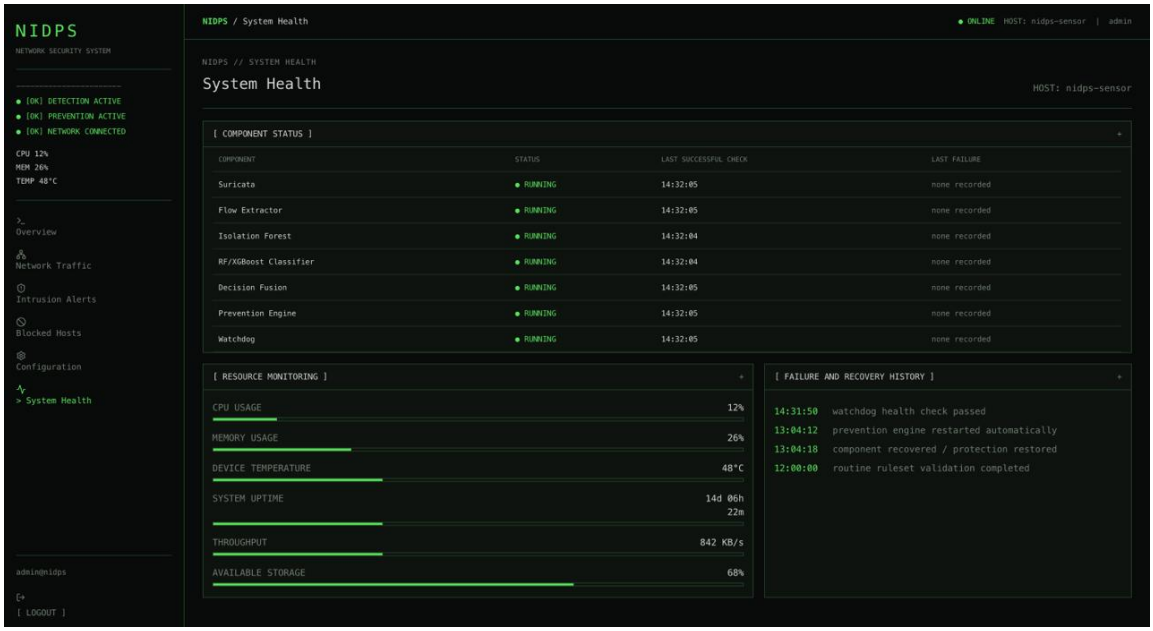


Figure 4.7: System Health screen.

The health screen shows whether every component is running and how the system has behaved over time, including periods in which it was not fully protecting the network. Covers: UC-11.

4.9 Database Design

The NIDPS stores derived data only: hosts, flow records for a short retention window, evidence, verdicts, blocks, the allowlist, model packages, calibration state, configuration, component health events and an audit log. A relational database is used because these records are strongly linked (for example, a block refers to a verdict, which refers to a flow and its evidence) and because the dashboard needs filtered, joined queries. SQLite is chosen because it runs in-process with no server, stores everything in one file and is sufficient for the write rate of a single home device. Raw packets and payloads are never stored. Live counters shown on the dashboard are kept in memory and on the event bus rather than in the database.

4.9.1 ER Diagram

Figure 4.8 shows the entity-relationship diagram. HOST is the central entity: a host has many flow records, evidence items, verdicts and blocks. Each flow record can have several evidence items (one per layer) and at most one final verdict, and a BLOCK verdict creates at most one block entry. Evidence produced by an ML layer refers to the model package

that produced it, and calibration states belong to the anomaly model. Administrative actions are linked to the admin user who performed them.

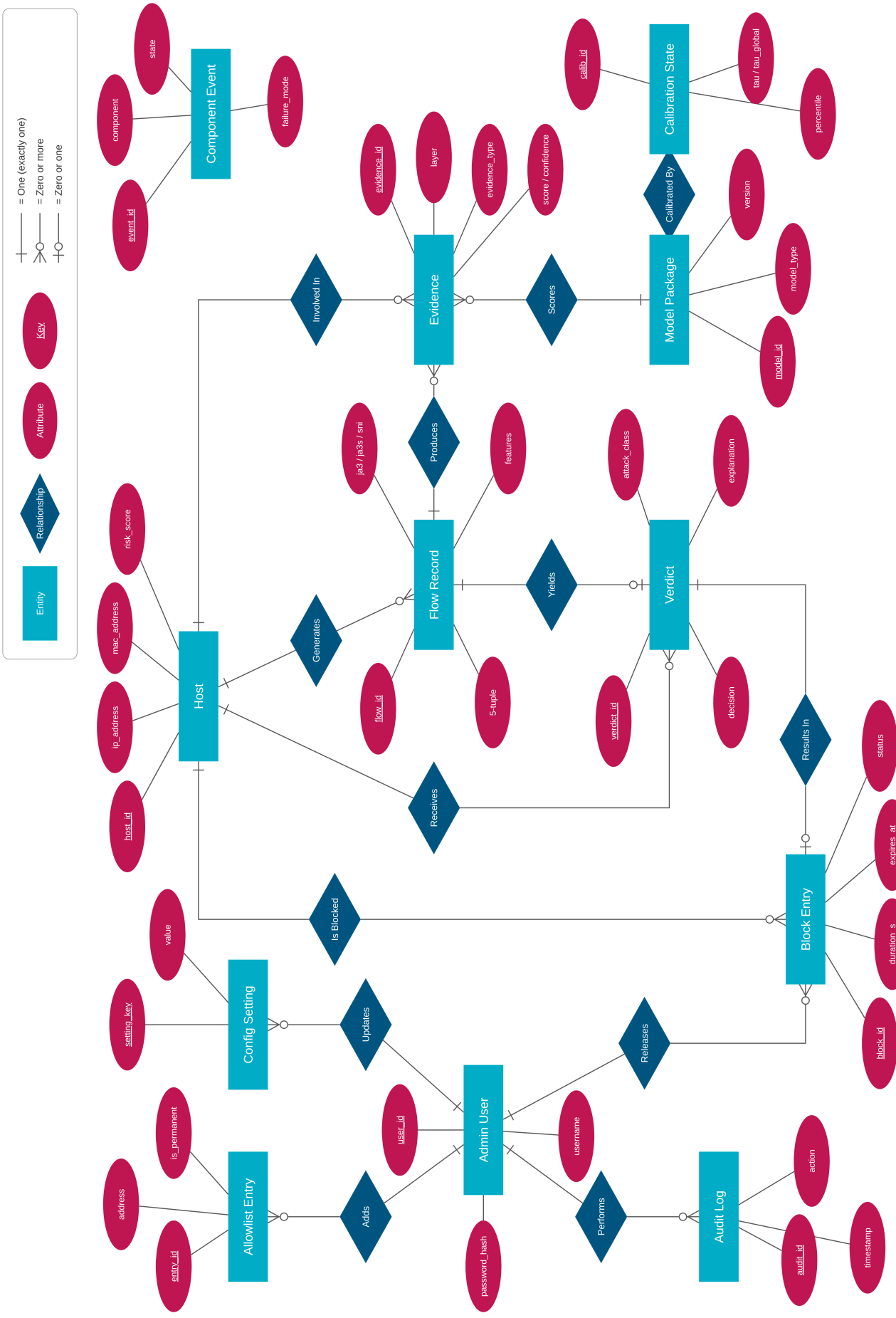


Figure 4.8: Entity-relationship diagram of the NIDPS database.

4.9.2 Data Dictionary

The tables below describe every entity. PK denotes a primary key and FK a foreign key. Timestamps are stored in UTC in ISO 8601 format.

Table 4.17: Data dictionary for ADMIN_USER.

Field	Type	Constraints	Description
user_id	INTEGER	PK	Unique identifier of the administrator
username	TEXT	UNIQUE, NOT NULL	Login name
password_hash	TEXT	NOT NULL	Salted password hash (e.g. Argon2/bcrypt)
failed_attempts	INTEGER	DEFAULT 0	Consecutive failed logins
locked_until	DATETIME	NULL	End of lock-out period, if locked
created_at / last_login	DATETIME		Account creation and last successful login

Table 4.18: Data dictionary for HOST.

Field	Type	Constraints	Description
host_id	INTEGER	PK	Unique identifier
ip_address	TEXT	NOT NULL	Current IPv4/IPv6 address
mac_address	TEXT	NULL	MAC address (internal hosts only)
hostname	TEXT	NULL	Name from DHCP lease or reverse DNS
is_internal	BOOLEAN	NOT NULL	True if the host is a client of the NIDPS
risk_score	REAL	DEFAULT 0	Current decaying evidence score S_h
first_seen / last_seen	DATETIME		First and most recent activity

Table 4.19: Data dictionary for FLOW_RECORD.

Field	Type	Constraints	Description
flow_id	INTEGER	PK	Unique identifier

Field	Type	Constraints	Description
host_id	INTEGER	FK → HOST	Internal host involved in the flow
src_ip, dst_ip, src_port, dst_port, protocol	TEXT / INTEGER	NOT NULL	5-tuple (stored for investigation, not used as model features)
start_time, end_time	DATETIME	NOT NULL	Flow or segment time span
segment_no	INTEGER	DEFAULT 0	Segment number for long flows
ja3, ja3s, sni	TEXT	NULL	TLS fingerprints and server name, if TLS
features	TEXT (JSON)	NOT NULL	78 flow features
extractor_version	TEXT	NOT NULL	Version of the flow extractor

Table 4.20: Data dictionary for EVIDENCE.

Field	Type	Constraints	Description
evidence_id	INTEGER	PK	Unique identifier
flow_id	INTEGER	FK → FLOW_RECORD, NULL	Related flow (NULL for ARP or host-level evidence)
host_id	INTEGER	FK → HOST	Host the evidence is about
layer	TEXT	CHECK IN (L1, ARP, L2, L3, HOST)	Detecting layer or host tracker
evidence_type	TEXT	NOT NULL	e.g. SIGNATURE, ANOMALY, CLASS, EXFIL, BEACON
signature_id	INTEGER	NULL	Suricata rule SID (Layer 1)
score	REAL	NULL	Anomaly or behaviour score
predicted_class, confidence	TEXT, REAL	NULL	Layer 3 class and probability
weight	REAL	NOT NULL	Weight w_v added to host score
model_id	INTEGER	FK → MODEL_PACKAGE, NULL	Model that produced ML evidence

Field	Type	Constraints	Description
created_at	DATETIME	NOT NULL	Time the evidence was produced

Table 4.21: Data dictionary for VERDICT.

Field	Type	Constraints	Description
verdict_id	INTEGER	PK	Unique identifier
flow_id	INTEGER	FK → FLOW_RECORD, NULL	Decided flow (NULL for host-score blocks)
host_id	INTEGER	FK → HOST	Host the verdict applies to
decision	TEXT	CHECK IN (PASS, ALERT, BLOCK)	Verdict
layers	TEXT	NOT NULL	Responsible layer(s), e.g. "L2+L3"
attack_class	TEXT	NULL	Unified attack class
explanation	TEXT	NOT NULL	Plain-language reason shown on the dashboard
reviewed	BOOLEAN	DEFAULT FALSE	Marked as reviewed by the administrator
created_at	DATETIME	NOT NULL	Time of the verdict

Table 4.22: Data dictionary for BLOCK_ENTRY.

Field	Type	Constraints	Description
block_id	INTEGER	PK	Unique identifier
host_id	INTEGER	FK → HOST	Blocked host
verdict_id	INTEGER	FK → VERDICT	Verdict that caused the block
offence_count	INTEGER	NOT NULL	Number of earlier blocks in 24 h (n)
duration_s	INTEGER	NOT NULL	Block duration D0 · 2^n, capped at D_max
start_time, expires_at	DATETIME	NOT NULL	Start and expiry of the block

Field	Type	Constraints	Description
status	TEXT	CHECK IN (ACTIVE, EXPIRED, RELEASED, SUPPRESSED)	Current state
released_by, released_at	INTEGER, DATETIME	FK → ADMIN_USER, NULL	Manual release details

Table 4.23: Data dictionary for ALLOWLIST_ENTRY.

Field	Type	Constraints	Description
entry_id	INTEGER	PK	Unique identifier
address	TEXT	UNIQUE, NOT NULL	IP or MAC address
description	TEXT		Reason, e.g. "Gateway"
is_permanent	BOOLEAN	DEFAULT FALSE	System entries that cannot be removed
added_by, added_at	INTEGER, DATETIME	FK → ADMIN_USER, NULL	Who added it and when

Table 4.24: Data dictionary for MODEL_PACKAGE.

Field	Type	Constraints	Description
model_id	INTEGER	PK	Unique identifier
model_type	TEXT	CHECK IN (IFOREST, RF, XGB)	Algorithm
version	TEXT	NOT NULL	Semantic version of the package
file_path	TEXT	NOT NULL	Location in the model registry
feature_list_hash, extractor_version	TEXT	NOT NULL	Schema compatibility information
class_list	TEXT (JSON)	NULL	Output classes (classifier only)
training_data, metrics	TEXT (JSON)		Datasets used and validation metrics

Field	Type	Constraints	Description
is_active	BOOLEAN	NOT NULL	True for the model currently in use
deployed_at	DATETIME	NOT NULL	Deployment time

Table 4.25: Data dictionary for CALIBRATION_STATE.

Field	Type	Constraints	Description
calib_id	INTEGER	PK	Unique identifier
model_id	INTEGER	FK → MODEL_PACKAGE	Anomaly model calibrated
tau, tau_global	REAL	NOT NULL	Calibrated and global thresholds
percentile	REAL	NOT NULL	Percentile p used
sample_count	INTEGER	NOT NULL	Number of clean flows used
u_max	INTEGER	NULL	Upload-size limit for exfiltration evidence
window_start, window_end, created_at	DATETIME	NOT NULL	Calibration window and run time

Table 4.26: Data dictionary for CONFIG_SETTING.

Field	Type	Constraints	Description
setting_key	TEXT	PK	e.g. failure_mode, active_timeout_s
value	TEXT	NOT NULL	Current value
default_value, min_value, max_value	TEXT		Default and allowed range
updated_by, updated_at	INTEGER, DATETIME	FK → ADMIN_USER, NULL	Last change

Table 4.27: Data dictionary for COMPONENT_EVENT.

Field	Type	Constraints	Description
event_id	INTEGER	PK	Unique identifier
component	TEXT	NOT NULL	Service name

Field	Type	Constraints	Description
state	TEXT	CHECK IN (RUNNING, DEGRADED, FAILED, RECOVERING)	State entered
failure_mode	TEXT	NULL	Mode applied during the failure
detected_at, recovered_at	DATETIME		Detection and recovery time
packets_dropped, packets_bypassed	INTEGER	DEFAULT 0	Traffic affected
details	TEXT		Cause and diagnostic information

Table 4.28: Data dictionary for AUDIT_LOG.

Field	Type	Constraints	Description
audit_id	INTEGER	PK	Unique identifier
user_id	INTEGER	FK → ADMIN_USER, NULL	Administrator (NULL for system actions)
action	TEXT	NOT NULL	e.g. LOGIN, BLOCK, RELEASE, CONFIG_CHANGE
target	TEXT		Affected object
old_value, new_value	TEXT	NULL	Values before and after a change
timestamp	DATETIME	NOT NULL	Time of the action

4.10 Risk Analysis

The main risks to the project are listed below with their category, likelihood, impact and mitigation. Technical risks concern whether the system can meet its requirements; resource, schedule and project risks concern whether the team can deliver it; legal and ethical risks concern how attack testing is carried out. The risk register is reviewed at every sprint review and updated as risks change.

Table 4.29: Risk analysis.

ID	Risk	Category	Likelihood	Impact	Mitigation
R1	Raspberry Pi cannot sustain household throughput with all layers enabled	Technical	Medium	High	Keep ML off the forwarding path; trim rule groups; degraded mode; measure saturation early and switch to a mini-PC if needed.
R2	Models trained on public datasets perform poorly on real home traffic (dataset shift)	Technical	High	High	Harmonised flow regeneration; per-network threshold calibration; Suricata as a baseline; self-collected test set; report results honestly.
R3	High false positive rate causes legitimate devices to be blocked	Technical / Business	Medium	High	Fusion requires strong or corroborated evidence; allowlist; rate limit; short first blocks; one-week alert-only soak test before enabling blocking.
R4	Wireless adapter or driver does not support stable AP mode or client isolation	Technical	Medium	Medium	Test hardware early; use a supported USB adapter; fall back to bridging lateral traffic where needed.
R5	Encrypted traffic and JA3 randomisation reduce detection coverage	Technical	High	Medium	Rely on flow metadata; combine JA3 with JA3S and SNI; isolate fingerprinting so JA4 can be adopted.
R6	Errors in public dataset labels or flow construction distort training	Technical	Medium	Medium	Regenerate flows from raw PCAPs; relabel from attack schedules; prefer corrected dataset versions where available.
R7	Large dataset PCAPs exceed available storage or memory for training	Resource	Medium	Medium	Process captures in chunks; sample benign traffic; use external storage or university lab machines.
R8	The NIDPS becomes a single point of failure and causes network outages	Operational	Low	High	Kernel-enforced fail-open default; watchdog; automatic restarts; timed blocks.
R9	An attacker poisons calibration or evades the models	Security	Low	High	Exclude flagged flows from calibration; bound threshold shift; no online retraining; signatures remain active.
R10	Attack experiments affect devices or networks outside the project	Legal / Ethical	Low	High	Run attacks only on an isolated lab network owned by the team; capture only the team's own traffic.

ID	Risk	Category	Likelihood	Impact	Mitigation
R11	Schedule slippage due to integration complexity of many services	Schedule	Medium	Medium	Incremental delivery of vertical slices; Layer 1 delivered first; sprint reviews with the supervisor.
R12	Hardware damage or failure (SD card corruption, overheating)	Resource	Medium	Medium	Use quality storage and cooling; keep image backups; keep a spare SD card.
R13	Team member unavailability	Project	Low	Medium	Shared repository and documentation; pair work on critical components; clear task ownership.

The two risks rated most serious are R2 and R3, because they determine whether the system is useful on a real network. Both are addressed by design (harmonised features, local calibration and cautious decision fusion) and are measured directly in the evaluation, so that their real effect is known rather than assumed.

Chapter 5

Proposed Approach and Methodology

This chapter presents the methodology through which the proposed Network Intrusion Detection and Prevention System (NIDPS) detects and blocks malicious traffic on a small network. The proposed approach is a hybrid detection framework: three complementary detection layers, a decision-fusion algorithm that turns their outputs into a single verdict, and an enforcement and recovery mechanism that acts on that verdict. The chapter begins with the datasets and the way they are harmonised so that models trained offline see the same features at runtime. It then walks through the backend algorithms that are not visible in the system design: flow construction, TLS fingerprinting, rule-based detection, Isolation Forest scoring with per-network threshold calibration, supervised attack classification, verdict fusion, automated blocking and the failure-handling state machine. The chapter closes with the evaluation protocol used to measure detection quality and the behaviour of the system on constrained hardware.

5.1 Overview of the Proposed Framework

The NIDPS runs on a single Linux host that acts as the network's wireless access point, so every packet between client devices and the ISP gateway passes through it. Each packet travels two paths at once. On the inline path, Suricata inspects packets as they are forwarded and can drop them immediately. On the analysis path, packets are grouped into bidirectional flows, and each completed flow is described by 78 statistical features plus TLS fingerprints (JA3/JA3S). These flow records are scored by an unsupervised Isolation Forest (Layer 2) and a supervised Random Forest/XGBoost classifier (Layer 3).
The three layers cover different threat classes:

- **Layer 1 – Signature detection (Suricata):** known attacks with stable signatures, such as SSH brute force, port scanning and DNS spoofing.
- **Layer 2 – Anomaly detection (Isolation Forest):** traffic that deviates from the network's normal profile, including zero-day behaviour, C2 beaconing and data exfiltration.
- **Layer 3 – Supervised classification (Random Forest/XGBoost):** assignment of suspicious flows to known attack families such as DDoS and lateral movement.

A decision-fusion engine combines the three outputs into one verdict per flow (pass, alert or block). Block verdicts are enforced through iptables and ipset. A watchdog supervises every component and applies the configured failure mode when one of them stops. Figure 5.1 summarises this pipeline.

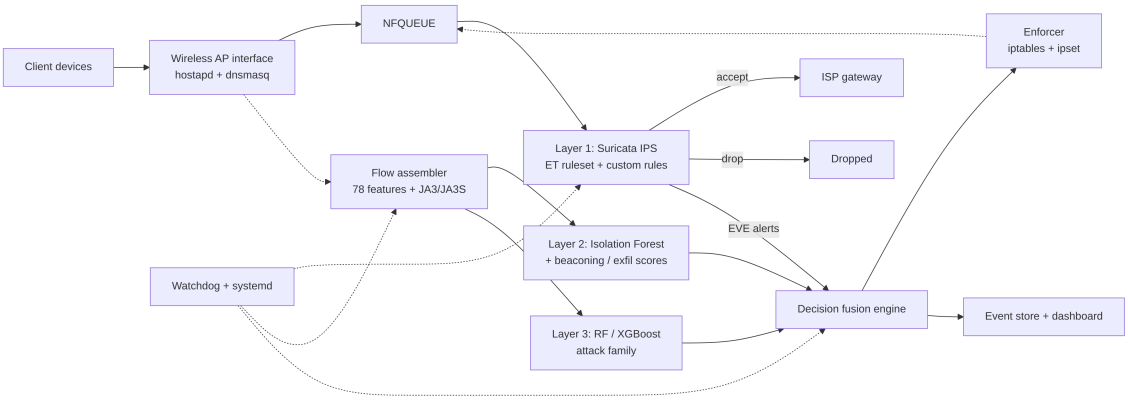


Figure 5.1: Proposed three-layer NIDPS pipeline.

The design separates what must be fast from what can be thorough. Suricata decides per packet and adds only microseconds of forwarding delay. The machine learning layers decide per flow, so their verdicts are applied to the remainder of a long flow and to all later traffic from the offending host.

5.2 Datasets

The machine learning layers are trained on three public intrusion detection benchmarks and evaluated additionally on traffic captured from our own deployment. The goal is not the highest score on one benchmark, but a model that still works when the training traffic and the deployment traffic differ.

5.2.1 Public Benchmark Datasets

Table 5.1 lists the public datasets used for training and cross-evaluation.

Table 5.1: Public benchmark datasets used for training and cross-evaluation.

Dataset	Source	Traffic profile	Attack families relevant to this project
CIC-IDS-2017 [16]	Canadian Institute for Cybersecurity (CIC)	5 days of enterprise-style traffic	Brute force (SSH/FTP), DoS/DDoS, port scan, botnet, infiltration
CSE-CIC-IDS2018 [17]	CIC with the Communications Security Establishment	Larger AWS-hosted network, 10 days	Brute force, DoS/DDoS, botnet, infiltration
CIC-IoT-2023 [18]	CIC	105 IoT devices in a smart-home topology	DDoS, recon/scan, spoofing, brute force, Mirai-style attacks

CIC-IoT-2023 is closest to the target environment because home networks are increasingly dominated by IoT devices. CIC-IDS-2017 and CSE-CIC-IDS2018 contribute a wider range of host-based attacks. Published analyses of CIC-IDS-2017 have reported labelling and flow-construction errors in the original CSV files [19], which is one more reason flows are regenerated from the raw captures (Section 5.2.3).

5.2.2 Self-Collected Real-World Dataset

A fourth dataset is captured on the deployed access point itself. It has two parts:

1. **Benign capture.** Normal household or lab traffic (browsing, streaming, video calls, IoT devices, software updates) is recorded over several days. Flows that Suricata flags during this period are removed before the capture is labelled benign.
2. **Attack capture.** Each in-scope attack is executed from a dedicated attacker machine in an isolated test network that we own, using standard penetration-testing tools. The start and end time, attacker address and target of each run are logged, so every flow in the window can be labelled automatically from these logs.

This dataset is used only for testing, never for training. It measures how well models trained on public data transfer to a real home network.

5.2.3 Feature Harmonisation Across Datasets

The public datasets do not share one feature format. The two CIC-IDS sets ship CICFlowMeter CSVs with about 80 columns whose names and rounding differ between releases, while CIC-IoT-2023 ships a different, smaller feature set computed over packet

windows. A model trained on one format cannot be applied to another, and none of them is guaranteed to match what the live system computes.
To remove this mismatch, the following procedure is applied:

1. Download the raw PCAP files of every dataset instead of the provided CSVs.
2. Run the same flow extractor that runs on the device (Section 5.3.2) over every PCAP. All datasets then share an identical 78-feature schema, with identical definitions, timeouts and units. The extractor runs with the same timeouts as the device ($T_a = 10\,\text{s}$, $T_i = 30\,\text{s}$), so a training record and a live record always cover the same span of traffic.
3. Re-label the regenerated flows by matching their 5-tuple and timestamps against each dataset's published attack schedule.
4. Map every dataset-specific label onto one unified attack taxonomy (Table 5.2).

The key advantage is that training-time and serving-time features are produced by the same code, so there is no train/serve skew.

Table 5.2: Unified attack taxonomy used for Layer 3.

Unified class	**Examples of original labels**	**Primary detecting layer**
Benign	BENIGN, BenignTraffic	–
Reconnaissance	PortScan, Recon-PortScan, Recon-OSScan	L1, L3
Brute force	SSH-Patator, FTP-Patator, DictionaryBruteForce	L1, L3
DoS / DDoS	DoS Hulk, DDoS LOIC, DDoS-SYN_Flood, DDoS-UDP_Flood	L1, L3
Botnet / C2	Bot, Mirai variants	L2, L3
Spoofing	DNS_Spoofing, MITM-ArpSpoofing	L1
Infiltration / lateral movement	Infiltration, internal scanning after compromise	L3

5.2.4 Pre-processing

The regenerated flow tables are cleaned before training:

- **Invalid values:** rows with infinite or undefined values (for example, rates computed over a zero-length flow) are dropped, and exact duplicate rows are removed.
- **Identifier removal:** source and destination IP addresses, ports, flow IDs and timestamps are removed from the feature vector. They identify the lab, not the attack,

and keeping them lets a model memorise the attacker's address instead of learning behaviour.

- **Constant features:** features with zero variance in the training set are removed.
- **Splitting:** within one dataset, the split into training and test sets is done by capture day rather than at random, because neighbouring flows of the same attack run are nearly identical and a random split leaks them into both sets.
- **Class imbalance:** class proportions are heavily skewed, in opposite directions: benign flows dominate CIC-IDS-2017, while attack flows dominate CIC-IoT-2023. Layer 3 uses class weights inversely proportional to class frequency; oversampling (SMOTE), if used, is applied only to the training fold.
- **Scaling:** tree-based models do not need feature scaling, so values are kept in their original units. This also keeps the online pipeline simple.
- **Isolation Forest training set:** Layer 2 is trained only on benign flows, since it models normal behaviour.

5.3 Traffic Acquisition and Flow Construction

All detection depends on how packets reach the detectors and how they are summarised. This section describes the inline packet path, the flow assembly algorithm and the extraction of TLS fingerprints from encrypted traffic.

5.3.1 Inline Packet Path

The host runs hostapd to expose a wireless network and dnsmasq to provide DHCP and DNS to clients. IP forwarding is enabled and the uplink interface performs NAT towards the ISP gateway. Forwarded traffic is diverted by an iptables rule in the FORWARD chain to an NFQUEUE, where Suricata runs in IPS mode. For each packet, Suricata returns an accept or drop verdict to the kernel.

A copy of every forwarded packet is also read by the flow assembler through a packet capture socket on the wireless interface. The flow assembler never holds packets back, so a slow ML layer cannot slow down forwarding.

One platform detail matters for lateral movement. By default, the wireless driver and hostapd relay frames between two wireless clients directly, without passing them up to the IP layer, so netfilter and Suricata never see them. To inspect client-to-client traffic, client isolation (`ap_isolate=1`) is enabled and intra-network traffic is routed through the host. Without this setting, lateral movement can only be detected when it crosses the AP's IP layer.

5.3.2 Flow Assembly Algorithm

A flow is the set of packets sharing the same 5-tuple (source IP, destination IP, source port, destination port, protocol) in either direction. The direction of the first packet defines "forward"; replies are "backward". Algorithm 5.1 shows how flows are built and closed.

Algorithm 5.1: Bidirectional flow assembly

Input: packet stream P; active timeout T_a (default 10 s); idle timeout T_i (default 30 s)

Output: completed flow records with 78 features

$FlowTable \leftarrow$ empty hash map keyed by canonical 5-tuple;
foreach *packet* $p \in P$ **do**
    $k \leftarrow \text{canonical}(p.\text{5tuple})$; `// same key for both directions`
    if $k \notin FlowTable$ **then**
        create flow f with $f.start \leftarrow p.time$ and $f.fwd \leftarrow$ direction of p;
        $FlowTable[k] \leftarrow f$;
    $f \leftarrow FlowTable[k]$;
    update the running statistics of f with p (direction, size, flags, time);
    if *p carries a TLS ClientHello or ServerHello* **then**
        attach JA3/JA3S to f (Algorithm 5.2);
    if *p has FIN in both directions or RST* **then**
        emit(f);
        remove k;
    else if $p.time - f.start \geq T_a$ **then**
        emit(f) as a partial record;
        reset the counters of f but keep k;
`// run periodically`
foreach *flow* f *with* $now - f.last_seen \geq T_i$ **do**
    emit(f);
    remove its key;

The active timeout lets a long-running connection, such as a slow exfiltration channel, be scored in segments while it is still open, instead of only when it closes. Each segment is emitted as its own flow record, after which its counters reset. The idle timeout reclaims memory from flows that stop without a FIN or RST. Table 5.3 gives the default values.

Table 5.3: Flow timeout parameters.

Parameter	**Default**	**Effect**
Active timeout T_a	10 s (configurable, e.g. 5–30 s)	An open flow is re-scored every T_a, which caps how long one connection can run unscored
Idle timeout T_i	30 s	A flow with no packets for T_i is closed and scored

Most flows, such as web requests, DNS lookups and scan probes, end in well under 5 seconds and are scored once, when they close. Only long-lived connections are segmented. T_a trades CPU for reaction time: halving it doubles the scoring work for long flows. The default of 10 s is a conservative starting point for low-cost hardware. T_a is a configuration parameter, so a deployment with spare CPU can lower it, for example to 5 s, to react faster. The evaluation measures both settings (Section 5.10.3).
The 78 features follow the CICFlowMeter definitions: packet and byte counts per direction, packet length statistics, inter-arrival time statistics, TCP flag counts, header lengths, bulk and sub-flow measures, and active/idle periods. To keep memory constant per flow, means and variances are updated incrementally with Welford's method (Equation 5.1) rather than by storing every packet:

$$\begin{aligned} \mu_n &= \mu_{n-1} + \frac{x_n - \mu_{n-1}}{n}, \\ M_n &= M_{n-1} + (x_n - \mu_{n-1})(x_n - \mu_n), \\ \sigma_n^2 &= \frac{M_n}{n-1}. \end{aligned} \tag{5.1}$$

Here x_n is the n-th observation (for example, a packet length), μ_n the running mean and M_n the running sum of squared deviations. Minimum and maximum are tracked directly.

5.3.3 TLS Fingerprint Extraction (JA3/JA3S)

Most traffic is encrypted, so payload cannot be inspected without decryption. Decrypting traffic on the AP would require installing a certificate on every client, would break certificate pinning in many apps and would cost far more CPU than the target hardware can spare. Instead, the system uses fields of the TLS handshake that are sent in clear text [20], as shown in Algorithm 5.2.

Algorithm 5.2: JA3/JA3S fingerprint

Input: TLS ClientHello (or ServerHello)

Output: 32-character MD5 fingerprint

read the TLS version, cipher suites, extension types, supported groups (elliptic curves) and EC point formats;

remove GREASE values (0x0a0a, 0x1a1a, . . . , 0xfafa) from every list;

join the decimal values of each list with "-";

$s_{JA3} \leftarrow$ Version,Ciphers,Extensions,Groups,PointFormats;

$s_{JA3S} \leftarrow$ Version,Cipher,Extensions ; `// ServerHello`

return *MD5(s)*;

The fingerprint identifies the client software that opened the connection, not the user. Malware and C2 frameworks often use TLS libraries with fingerprints that differ from common browsers and operating systems. The pair (JA3, JA3S) is attached to the flow record and used by Layer 2 in two ways: as a rarity signal (how many local hosts have ever used this fingerprint) and as part of the key for beacon detection (Section 5.5.4). Suricata's TLS parser can emit the same hashes in its EVE logs, which lets Layer 1 rules match known-bad fingerprints directly.

5.4 Layer 1: Signature-Based Detection

Layer 1 catches attacks that have a known, stable pattern. It is the only layer that acts on individual packets, so it is the first line of prevention.

5.4.1 Suricata Configuration

Suricata [21] runs in IPS mode on the NFQUEUE with the Emerging Threats Open ruleset. Rule categories that do not apply to a home network (for example, SCADA protocols) are disabled to save CPU. Each rule carries an action (alert or drop) and a severity from 1 (highest) to 3. Rules with severity 1 that target in-scope attacks are set to drop; the rest only alert and are passed on to decision fusion. Every alert is written to Suricata's EVE JSON output, which the alert ingestor reads over a Unix socket.

5.4.2 Custom Rules for In-Scope Attacks

The ET ruleset matches known payloads well, but several in-scope attacks are defined by rate rather than content. These are covered by custom threshold rules, which count matching events per source address within a time window (Table 5.4).

Table 5.4: Custom Layer 1 rules for the in-scope attacks.

Attack	**Detection logic**	**Example threshold (tunable)**
SSH brute force	New TCP connections (SYN) to port 22 from one source	≥ 5 in 60 s
Port scan	SYN packets from one source to many distinct destination ports	≥ 20 ports in 10 s
DNS spoofing	DNS responses to clients whose source is not the AP's own resolver	any occurrence
Known C2 TLS client	JA3 hash on a blocklist of published malicious fingerprints	any occurrence
SYN flood (DoS)	SYN packets to one destination without completed handshakes	rate above a calibrated limit

The thresholds are starting values that are tuned during evaluation. The DNS spoofing rule relies on the AP design. Clients receive the AP's address as their only DNS server through DHCP, so a legitimate DNS answer always comes from the AP. Any DNS response arriving at a client from another address is forged or comes from a rogue resolver.

5.4.3 ARP Spoofing Monitor

Suricata inspects IP traffic and does not match rules against ARP frames, which sit below the IP layer. ARP spoofing is therefore detected by a small companion sensor that belongs to Layer 1 and reports into the same alert stream. It uses the DHCP lease table of dnsmasq as the ground truth for which MAC address owns which IP address (Algorithm 5.3).

Algorithm 5.3: ARP consistency check

Input: ARP frames on the wireless interface; DHCP lease table L (IP $\rightarrow$ MAC)
Output: ARP spoofing alerts

$Bindings \leftarrow L$; `// trusted IP-to-MAC map`
foreach *ARP reply or gratuitous ARP a* **do**
  if $a.sender_ip \in Bindings$ ***and*** $Bindings[a.sender_ip] \neq a.sender_mac$ **then**
    raise alert ARP_SPOOF (attacker $a.sender_mac$, victim $a.sender_ip$);
  if $a.sender_ip = gateway_ip$ ***and*** $a.sender_mac \neq AP_mac$ **then**
    raise alert GATEWAY_IMPERSONATION with severity 1;
  if *more than N unsolicited replies from one MAC within window W* **then**
    raise alert ARP_FLOOD;
on every DHCP lease change, refresh *Bindings* from L;

Impersonating the gateway is the classic step before a man-in-the-middle attack, so that check raises the highest severity.

5.5 Layer 2: Unsupervised Anomaly Detection

Layer 2 flags traffic that looks unlike the network's normal behaviour, without needing a label for the attack. It covers what signatures miss: new attacks, variants that change their signature, C2 beaconing and data exfiltration.

5.5.1 Isolation Forest

Isolation Forest [22] is based on a simple observation: anomalies are few and different, so they are easier to separate from the rest of the data than normal points are. The model builds an ensemble of random binary trees. Each tree is grown on a random subsample of ψ flows by repeatedly choosing a random feature and a random split value between that feature's minimum and maximum, until every point is isolated or a height limit of $\lceil \log_2 \psi \rceil$ is reached.

An anomalous flow is isolated after few splits, so its path length $h(x)$ from root to leaf is short. Averaging over all trees gives $E[h(x)]$, which is normalised by the average path length of an unsuccessful search in a binary search tree of ψ points:

$$c(\psi) = 2H(\psi - 1) - \frac{2(\psi - 1)}{\psi}, \qquad H(i) \approx \ln(i) + 0.5772 \tag{5.2}$$

$$s(x, \psi) = 2^{-\frac{E[h(x)]}{c(\psi)}} \tag{5.3}$$

The anomaly score s in Equation 5.3 lies between 0 and 1. Scores close to 1 indicate clear anomalies; scores well below 0.5 indicate normal points. The model uses $t = 100$ trees and $\psi = 256$, the values recommended by the original authors, and is trained only on benign flows from the harmonised datasets.

Isolation Forest suits the target hardware because training and scoring are cheap. Scoring a flow costs t tree walks of at most $\lceil \log_2 \psi \rceil = 8$ steps each, regardless of how much training data was used.

5.5.2 Adaptive Threshold Calibration

The model outputs a continuous score, so a threshold τ is needed to turn it into a decision. A threshold taken from benchmark traffic does not fit every home network: a household with game consoles and security cameras produces traffic that a benchmark lab never saw, which shows up as false positives. The calibration scheme in Algorithm 5.4 adjusts only the threshold to each deployment; it does not retrain the model.

Algorithm 5.4: Per-network threshold calibration

Input: trained Isolation Forest M; global threshold τ_g; calibration window W (e.g. 24–72 h); percentile p (e.g. 99.5); minimum sample count N_{min}; maximum shift δ

Output: calibration state $\{\tau, \text{timestamp}, n\}$

$S \leftarrow \emptyset$;
foreach *flow x completed during W* **do**
  if *x matched any Suricata alert* ***or*** *came from a blocked host* **then**
    skip x ; `// keep the baseline clean`
  add M.score(x) to S;
if $|S| < N_{min}$ **then**
  $\tau \leftarrow \tau_g$; `// too little local data: fall back`
else
  $\tau_l \leftarrow p$-th percentile of S;
  $\tau \leftarrow \text{clip}(\tau_l,\ \tau_g - \delta,\ \tau_g + \delta)$;
store $\{\tau, now, |S|\}$ as calibration metadata;
repeat periodically (e.g. weekly) as the network's traffic changes;

Two safeguards protect the calibration. The exclusion step leaves out flows that Layer 1 already considers malicious. The clipping step limits how far the local threshold may move from the global one, so an attacker who is active during calibration cannot raise the threshold far enough to hide.

The percentile p directly sets the expected false positive rate on the calibration traffic: with $p = 99.5$, about 0.5% of benign local flows would exceed τ. This makes the trade-off between missed attacks and alert volume an explicit, tunable parameter.

5.5.3 Data Exfiltration Score

Exfiltration is a property of a host over time, not of one flow. For each internal host h, the system keeps an exponentially weighted moving average and variance of its outbound bytes per time bucket b_t (60 s by default):

$$\bar{b}_t = \alpha\, b_t + (1-\alpha)\, \bar{b}_{t-1}, \qquad \sigma_t^2 = \alpha\, (b_t - \bar{b}_{t-1})^2 + (1-\alpha)\, \sigma_{t-1}^2 \tag{5.4}$$

A bucket is flagged when b_t exceeds the host's baseline by k standard deviations ($k = 4$ as a starting value) and the outbound-to-inbound byte ratio of the host's flows is also unusually high. The second condition prevents large uploads that are part of normal two-way activity, such as video calls, from triggering the alert.

Bucket statistics react only once per bucket, so a second, faster check runs on every active-timeout segment. If one segment carries more than U_{max} bytes outbound to a destination the host has never contacted before, a high-weight evidence item is raised immediately. U_{max} is set during calibration (Algorithm 5.4) as a high percentile, for example the 99.9th, of

segment upload sizes in local benign traffic. When this evidence leads to a block, leakage over a single connection is bounded by roughly the host's upload rate $\times$ (T_a + scoring delay). At 20 Mbps and the default $T_a = 10$ s, that is about 25 MB.

5.5.4 C2 Beaconing Score

Infected devices typically contact their command-and-control server at regular intervals. Flows are grouped by the key (internal host, destination IP or TLS server name, JA3). For each group with at least m connections in the observation window, the inter-arrival times $\Delta t_1, \ldots, \Delta t_k$ between consecutive connections are computed, along with their coefficient of variation:

$$CV = \frac{\sigma_{\Delta t}}{\mu_{\Delta t}} \tag{5.5}$$

Human-driven traffic is irregular (high CV), while beacons are close to periodic (low CV), even with added random jitter. The beacon score combines three signals: a low CV, a low variance in bytes per connection, and a rare JA3 fingerprint on the local network. A group is reported when all three point the same way, which keeps legitimate periodic traffic, such as software update checks from common browsers, from being flagged.

5.6 Layer 3: Supervised Attack Classification

Layer 3 assigns each flow to one of the unified classes of Table 5.2. Where Layer 2 answers "is this unusual?", Layer 3 answers "which known kind of attack does this look like?". The class label decides the response and makes alerts readable on the dashboard. Two tree ensembles are trained and compared, and the better one under the hardware budget is deployed.

5.6.1 Random Forest

Random Forest [23] trains many decision trees, each on a bootstrap sample of the training flows, and at each split considers only a random subset of $\sqrt{d}$ of the d features. Splits are chosen to minimise Gini impurity:

$$G(S) = 1 - \sum_{k=1}^{K} p_k^2 \tag{5.6}$$

where p_k is the share of class k among the flows S at a node. The forest predicts by averaging the class probabilities of all trees. Bootstrapping and feature subsampling make the trees disagree in useful ways, so the ensemble is less prone to overfitting than a single tree. This matters because the model is trained on lab traffic and deployed on a different network.

5.6.2 XGBoost

XGBoost [24] builds trees one after another, each new tree correcting the errors of the current ensemble. It minimises a regularised objective:

$$\mathscr{L} = \sum_i l(y_i, \hat{y}_i) + \sum_m \Omega(f_m), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2 \tag{5.7}$$

Here l is the multi-class log loss, T is the number of leaves of tree f and w its leaf weights. The γ and λ terms penalise large, complex trees. XGBoost is usually more accurate than Random Forest on tabular data, but it is more sensitive to hyperparameters and can fit dataset-specific artefacts more closely, which the cross-dataset evaluation (Section 5.10.2) is designed to expose.

5.6.3 Training and Model Selection

1. Train on the harmonised flows with identifiers removed (Section 5.2.4), using class weights to counter imbalance.
2. Tune hyperparameters (number of trees, maximum depth, learning rate for XGBoost, minimum samples per leaf) with stratified 5-fold cross-validation on the training days only.
3. Select by macro-averaged F1-score rather than accuracy. With heavily skewed classes, a model that always predicts the majority class can still look accurate while missing the rare classes entirely.
4. Apply a hardware budget: among models within one F1 point of the best, choose the one with the lowest per-flow inference time and memory on the target device.
5. Export the chosen model with its feature list and version number, so the runtime refuses a model whose features do not match the extractor.

5.6.4 Confidence Handling

The classifier always outputs a probability vector p over the K classes. The predicted class is $\hat{y} = \arg\max p$, with confidence $c = \max p$. Low confidence means the flow does not fit any class well, which often happens with attacks the model never saw. Instead of forcing a label, flows with c below a confidence threshold θ_c are labelled "unknown" and passed to decision fusion with their Layer 2 score. This way, Layer 3 refines detections but does not overrule a strong anomaly signal.

5.7 Decision Fusion

The three layers can disagree, and acting on any single one would either block too much (anomaly scores alone are noisy) or too little (signatures alone miss new attacks). The

decision-fusion engine combines their evidence per flow and per host. Its main principle is that strong evidence from one layer is enough to block, while weak evidence must be confirmed by a second layer or must accumulate over time.

5.7.1 Verdict Algorithm

Algorithm 5.5 gives the per-flow verdict, and Figure 5.2 shows the same logic as a flowchart.

Algorithm 5.5: Per-flow verdict

Input: flow f; Suricata alerts $A(f)$; anomaly score $s(f)$; threshold τ; class probabilities $p(f)$; thresholds $\theta_{block} > \theta_{alert} > \theta_c$

Output: verdict in {PASS, ALERT, BLOCK} and the attributed layer(s)

if $f.src \in$ *allowlist* **then return** *PASS*;
if $A(f)$ *contains a drop rule or severity 1* **then return** *BLOCK, layer L1*;
$y \leftarrow \arg\max p(f)$; $\quad c \leftarrow \max p(f)$;
if $y \neq$ *Benign* ***and*** $c \geq \theta_{block}$ **then return** *BLOCK, layer L3*;
if $s(f) \geq \tau$ ***and*** $y \neq$ *Benign* ***and*** $c \geq \theta_{alert}$ **then return** *BLOCK, layers L2+L3*;
if $s(f) \geq \tau$ ***or*** $A(f) \neq \emptyset$ ***or*** $(y \neq$ *Benign* ***and*** $c \geq \theta_{alert})$ **then**
    add evidence to the host score of $f.src$ (Section 5.7.2);
    return *ALERT*;
return *PASS*;

A confident known attack is blocked on Layer 3 alone, while a less confident classification is blocked only when Layer 2 agrees.

5.7.2 Host Evidence Accumulation

Some attacks, such as a slow port scan or low-volume exfiltration, never produce one strong signal. Their individual flows only trigger alerts. The host score S_h collects this weak evidence and lets it fade over time, so that a single stray alert does not lead to a block but a steady stream of alerts does:

$$S_h(t) = S_h(t_{prev})\, e^{-\lambda (t - t_{prev})} + w_v \tag{5.8}$$

Here w_v is the weight of the new alert (for example, higher for a Suricata alert than for an anomaly score just above τ) and λ is a decay rate. With a half-life of 10 minutes, $\lambda = \ln 2/600 \text{ s}^{-1}$. When S_h reaches the block threshold Θ, the host is blocked. The same score is shown on the dashboard as the host's current risk level.

5.7.3 Timing of ML Verdicts

Layers 2 and 3 score completed flows or active-timeout segments (Algorithm 5.1), so their verdict comes after part of the traffic has passed. Their blocks therefore apply to the rest of a long flow and to all later traffic from that source. This is acceptable for the attacks they

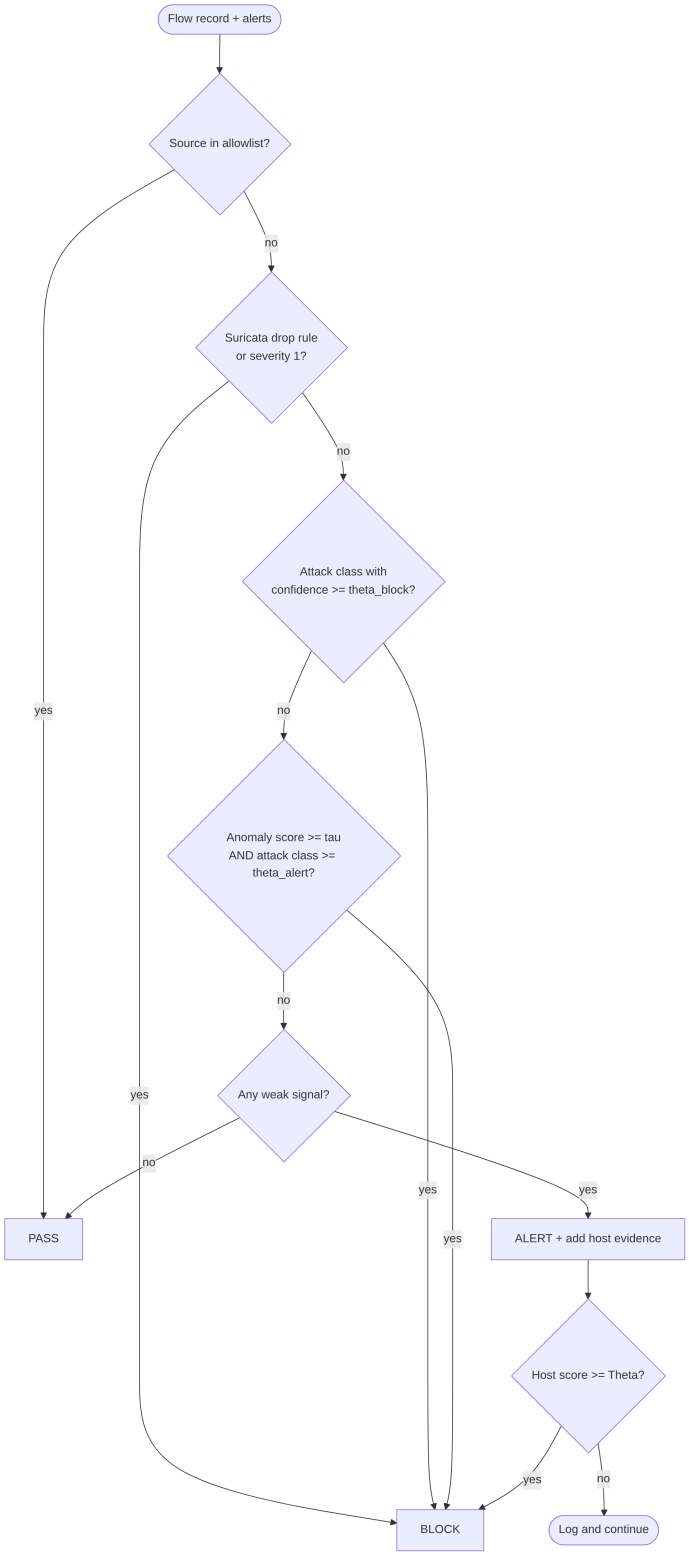


Figure 5.2: Decision-fusion flow for one flow record.

target: beaconing, exfiltration, DDoS and lateral movement all unfold over many flows. Attacks that must be stopped at the first packet are the responsibility of Layer 1.
With the default $T_a = 10\,\text{s}$, the worst-case delay for an ML verdict on a long flow is about 10 seconds plus scoring time, which is milliseconds for tree models. The data that can leave over one connection before a block is therefore roughly the upload rate $\times$ that delay. This exposure is measured directly in the live exfiltration tests as "bytes leaked before block" (Table 5.7), rather than estimated.

5.8 Automated Prevention

A BLOCK verdict is turned into a firewall entry by the enforcer. The enforcer must stop attacks quickly without ever locking the owner out of their own network, so blocking follows a strict procedure.

5.8.1 Blocking Mechanism

Blocked addresses are stored in an ipset with a per-entry timeout, referenced by a single iptables rule placed before the NFQUEUE rule in the FORWARD chain (and in INPUT, to protect the AP itself). Adding or removing an address changes only the set, not the rule list, so a block takes effect immediately and costs the same regardless of how many addresses are blocked. Entries expire automatically, so a block never outlives its intended duration even if the enforcer crashes.

5.8.2 Blocking Procedure

Algorithm 5.6 gives the procedure followed for every BLOCK verdict.

Algorithm 5.6: Enforce a BLOCK verdict

Input: offending address a; attack class y; attributed layer(s)
Output: ipset entry and audit record

if $a \in$ *allowlist (AP, gateway, DNS resolver, admin devices)* **then**
    log "block suppressed";
    return;
if *blocks issued in the last minute* $\geq R_{max}$ **then**
    downgrade to ALERT;
    raise a "block rate limit" warning;
    return;
$n \leftarrow$ number of earlier blocks of a within 24 h;
$duration \leftarrow \min(D_0 \cdot 2^n,\ D_{max})$; // e.g. $D_0 = 5\,\text{min}$, $D_{max} = 24\,\text{h}$
add a to the ipset with timeout $duration$;
write audit record $\{a, y, \text{layers}, duration, \text{evidence}, \text{time}\}$;
push the event to the dashboard;

The allowlist check prevents the system from cutting off its own infrastructure, even when an attacker forges packets from those addresses. The rate limit stops a flood of false positives, or an attacker spoofing many sources, from turning the NIDPS into a denial-of-service tool against its own users. Exponential back-off means a first offence is blocked briefly, in case it was a false positive, while repeat offenders are blocked for longer. Every block can be lifted manually from the dashboard.

5.9 Fault Tolerance and Failure Handling

An inline security device is itself a single point of failure: if Suricata crashes while traffic is queued to it, the whole network can go down. The NIDPS therefore defines exactly what happens when each component fails, and lets the owner choose between availability and security.

5.9.1 Health Monitoring

Every component (Suricata, flow assembler, ML scorer, decision engine, enforcer) runs as a separate systemd service. Each one sends a heartbeat to the watchdog every second, along with load indicators such as queue depth and processing lag. A component is declared failed after $k = 3$ missed heartbeats or if its process exits. systemd restarts failed services automatically, and its own watchdog timer restarts services that are running but frozen.

5.9.2 Failure Modes

Table 5.5 lists the configurable failure modes.

Table 5.5: Configurable failure modes.

Mode	When Layer 1 fails	When an ML layer fails	Suited to
Fail-open	Packets bypass the queue and are forwarded uninspected	Remaining layers continue; affected flows are logged as unscored	Homes where internet access matters most
Fail-closed	Forwarded traffic is dropped until recovery; admin access to the AP stays open	Flows from unknown hosts are held at ALERT and new block decisions rely on Layer 1 only	Small offices handling sensitive data
Degraded (load-shedding)	Non-essential rule groups are disabled to reduce load	Only a sample of flows (1 in k) is scored, prioritising new and high-risk hosts	Temporary overload rather than a crash

Fail-open is implemented in the kernel rather than in software: the NFQUEUE rule is created with the queue-bypass option, so when no process is attached to the queue, the kernel accepts packets instead of dropping them. Fail-closed omits this option. Because the kernel enforces the mode, it holds even when the watchdog itself has crashed.

5.9.3 State Machine

Figure 5.3 shows the states each supervised component moves through.

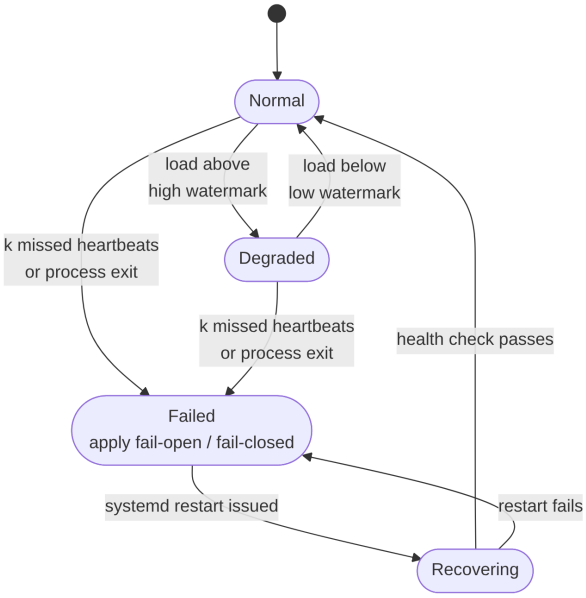


Figure 5.3: Watchdog state machine for each supervised component.

Two thresholds (high and low watermarks) are used for entering and leaving the degraded state. Without this gap, a load hovering around one value would cause the system to switch modes back and forth many times per second. During every failure, the watchdog records when it was detected, which mode was applied, when service resumed and how many packets were dropped or bypassed. These records produce the recovery measurements reported in the evaluation.

5.10 Evaluation Methodology

The system is evaluated on two questions: how well it detects attacks, and whether it can do so inline on constrained hardware without harming the network.

5.10.1 Detection Metrics

For every layer and for the fused system, precision, recall, F1-score and false positive rate (FPR) are reported per class and macro-averaged:

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP+FP}, & \text{Recall} &= \frac{TP}{TP+FN}, \\ F_1 &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, & \text{FPR} &= \frac{FP}{FP+TN} \end{aligned} \tag{5.9}$$

False positive rate receives particular attention. On a home network producing thousands of flows per hour, even a 1% false positive rate means dozens of wrong alerts or blocks every hour. For Layer 2, the area under the precision-recall curve is also reported, since it summarises performance across all thresholds and is more informative than ROC-AUC when attacks are rare.

5.10.2 Cross-Dataset Generalisation

Each model is trained on one dataset and tested on every other dataset, giving the matrix in Table 5.6. The diagonal (train and test on the same dataset, split by day) shows in-distribution performance. The off-diagonal cells show how much performance drops when the network changes, which is the realistic situation for a deployed device. Finally, models trained on all three public datasets combined are tested on the self-collected dataset.

Table 5.6: Cross-dataset evaluation matrix of macro F1-scores, to be filled from experiments.

Train / Test	**CIC-IDS-2017**	**CSE-CIC-IDS2018**	**CIC-IoT-2023**	**Self-collected**
CIC-IDS-2017	in-distribution			
CSE-CIC-IDS2018		in-distribution		
CIC-IoT-2023			in-distribution	
All three combined				

Only classes that exist in both the training and the test dataset are scored in each cell. The adaptive threshold of Section 5.5.2 is evaluated by comparing the false positive rate on the self-collected benign traffic with the global threshold and with the calibrated threshold.

5.10.3 System Performance on Constrained Hardware

Table 5.7 defines the system-level measures and how each is obtained.

Table 5.7: System performance measures.

Measure	**Definition**	**How it is measured**
Forwarding latency	Extra round-trip time added by the NIDPS	Ping RTT through the AP with the NIDPS off, then with each layer enabled in turn
Throughput	Sustained forwarding rate	iperf3 TCP and UDP tests through the AP, layers enabled incrementally
Detection latency	Time from the first attack packet to the alert	Attack start logged on the attacker machine vs. alert timestamp
Time to block	Time from the first attack packet to the ipset entry	Same, against the audit record
Bytes leaked before block	Data the attacker transferred before the block took effect	Bytes received at the lab exfiltration server before the block, repeated at $T_a = 5$ s and 10 s
Resource use	CPU, memory and temperature of the host	Sampled every second during each test, at $T_a = 5$ s and 10 s
Recovery behaviour	Time to detect a failure, time to restore service, packets lost or bypassed	Components killed deliberately under load, in each failure mode

Every attack in the scope is replayed live against test devices on the isolated lab network while background benign traffic runs. This end-to-end test checks the complete chain, from detection through fusion to blocking, rather than each model in isolation.

5.11 Conclusion

This chapter presented the proposed hybrid framework and the backend algorithms behind it. Suricata and an ARP consistency check provide fast, packet-level detection of known attacks. Isolation Forest, with a per-network threshold calibration, detects unusual behaviour, supported by host-level scores for exfiltration and C2 beaconing. A Random Forest or XGBoost classifier assigns flows to known attack families. Encrypted traffic is analysed through JA3/JA3S fingerprints and flow statistics instead of decryption, and all datasets are regenerated with the device's own flow extractor so training and runtime features match. A decision-fusion algorithm with host evidence accumulation turns these signals into verdicts, an enforcer applies them through timed ipset entries with safeguards against self-inflicted outages, and a watchdog applies a configurable failure mode backed by the kernel. The evaluation protocol measures both detection quality across datasets and the system's la-

tency, throughput and recovery on real hardware. The next chapter describes how these algorithms are organised into components and how those components interact.

Chapter 6

High-Level and Low-Level Design

This chapter describes how the methodology of Chapter 5 is realised as software. It moves from the high level to the low level: first an overview of the system and the considerations that shaped its design, then the architecture of the whole system and of each subsystem, the strategies behind the main architectural decisions, the class diagram, and finally the policies and tactics that govern implementation, testing and maintenance.

6.1 System Overview

The NIDPS is a self-contained security appliance built from software on a single Linux host. The host replaces the role of the home router's wireless network: clients connect to the NIDPS access point, and the NIDPS routes their traffic, through its own NAT, to the existing ISP router, which keeps handling the internet connection. No change is needed on client devices or on the ISP router.
Functionally, the system does five things:

1. **Provides network access:** a wireless access point with DHCP, DNS and routing.
2. **Inspects traffic:** inline signature inspection of every packet, plus flow-level analysis of all traffic, including encrypted traffic through metadata.
3. **Decides:** fuses signature, anomaly and classification results into a per-flow verdict and a per-host risk score.
4. **Responds:** blocks offending addresses automatically for a limited, escalating time.
5. **Reports and stays available:** shows live statistics and alerts on a web dashboard, and keeps the network in a defined state when a component fails.

The system is organised as a set of small, independent services connected by a local event bus, rather than as one large program. The packet-forwarding path is kept as short as possible (kernel plus Suricata), and everything slower, such as machine learning, the database and the dashboard, sits beside that path instead of in it. Sections 6.3 and 6.4 explain this choice in detail.

6.2 Design Considerations

This section records the assumptions, constraints, goals and development methods that shaped the design.

6.2.1 Assumptions and Dependencies

- **Hardware:** the host has at least a quad-core 64-bit ARM or x86 CPU, 4 GB of RAM, a wireless adapter that supports AP mode and an Ethernet port for the uplink. A Raspberry Pi 4/5 class board or a small x86 mini-PC meets this. A USB Wi-Fi adapter is assumed when the built-in radio does not support AP mode reliably.
- **Operating system:** a Debian-based Linux distribution with systemd, iptables (nftables backend accepted), ipset and NFQUEUE support in the kernel.
- **Network position:** the NIDPS is the only wireless network clients use. Devices that connect directly to the ISP router by cable or by the router's own Wi-Fi are not protected; disabling the ISP router's Wi-Fi is part of setup.
- **Third-party software:** Suricata 7 or later with the Emerging Threats Open ruleset, hostapd, dnsmasq, and Python 3.11+ with scikit-learn and XGBoost. Ruleset updates require periodic internet access.
- **End users:** the owner is not a security expert. Defaults must be safe without tuning, and the dashboard must explain alerts in plain language.
- **Trust:** the NIDPS host itself and the admin's devices are trusted. The ISP router is assumed to be functional but is not trusted to provide any security.
- **Likely changes:** new attack types and rules will be added, models will be retrained and the TLS fingerprinting method may move from JA3 to its successor JA4 as client libraries randomise extension order. The design isolates each of these behind its own component.

6.2.2 General Constraints

Table 6.1 lists the global constraints and their impact on the design.

Table 6.1: General constraints and their impact on the design.

Constraint	Source	Impact on the design
Limited CPU and memory	Low-cost target hardware	Tree-based models instead of deep networks; per-flow rather than per-packet ML; incremental feature statistics; rule groups trimmed to the home-network scope
Inline position	The NIDPS forwards all traffic	Forwarding path kept to kernel + Suricata; ML runs off-path; failure mode enforced in the kernel
Encrypted traffic	Most traffic uses TLS/QUIC	No payload inspection for ML; detection from flow metadata and JA3/JA3S only
Privacy	Household traffic is personal	No decryption; no payload storage; only flow metadata and alerts are kept, with a retention limit
Availability	A home network must not go down	Configurable fail-open default; block rate limits; allowlist; ipset entries that expire
Interoperability	Training data and live traffic must match	One flow extractor used offline and online; model files carry their feature schema
Security of the device	The NIDPS is itself a target	Dashboard bound to the LAN only, authenticated, management traffic allowlisted; services run without root where possible
Testability	Attacks cannot be run on real users	Isolated lab network and attacker machine for all attack tests

6.2.3 Goals and Guidelines

1. **Never make the network worse than a plain router.** Every design choice is checked against this. It is why fail-open is the default, why blocks expire and why the number of blocks per minute is capped.
2. **Keep the fast path short (KISS).** Only the kernel and Suricata sit in the forwarding path. Anything that can be done after the fact is moved off it.
3. **Defence in depth.** No single detector is trusted to catch everything; each layer covers a weakness of the others (Section 5.1).
4. **Speed over memory where the two conflict, within a fixed budget.** Models and rule sets are loaded fully in memory to avoid disk access during inspection, but their size is capped to fit the hardware.

5. **Explainable decisions.** Every block records which layer decided, why, and with what evidence, so the owner can trust or override it.
6. **Same code for training and serving.** Features are computed by one implementation to rule out train/serve skew.
7. **Configurable, safe defaults.** Thresholds, failure mode and block durations are configurable, but the defaults work without tuning.

6.2.4 Development Methods

The project follows an iterative, incremental approach in short sprints of two weeks, aligned with the FYP milestones. Each increment delivers a working vertical slice: first the access point with Suricata (Layer 1, completed), then flow extraction and Layer 2, then Layer 3, then fusion and enforcement, and finally the watchdog and dashboard. This order means the system protects the network from the first increment onwards and every later layer can be measured against a working baseline.
The machine learning work follows the CRISP-DM cycle (data understanding, preparation, modelling, evaluation, deployment), because model quality depends on several rounds of data cleaning and re-evaluation. A plain waterfall model was considered and rejected: detection performance on real traffic cannot be known in advance, so requirements such as thresholds and feature sets must be allowed to change as results come in.

6.3 System Architecture

The system is decomposed into five runtime subsystems on the NIDPS host and one offline subsystem used for training. Figure 6.1 shows the external architecture (how the NIDPS sits between clients, the ISP router and the administrator) together with the internal subsystems and their main data flows.
The responsibilities are partitioned as shown in Table 6.2.

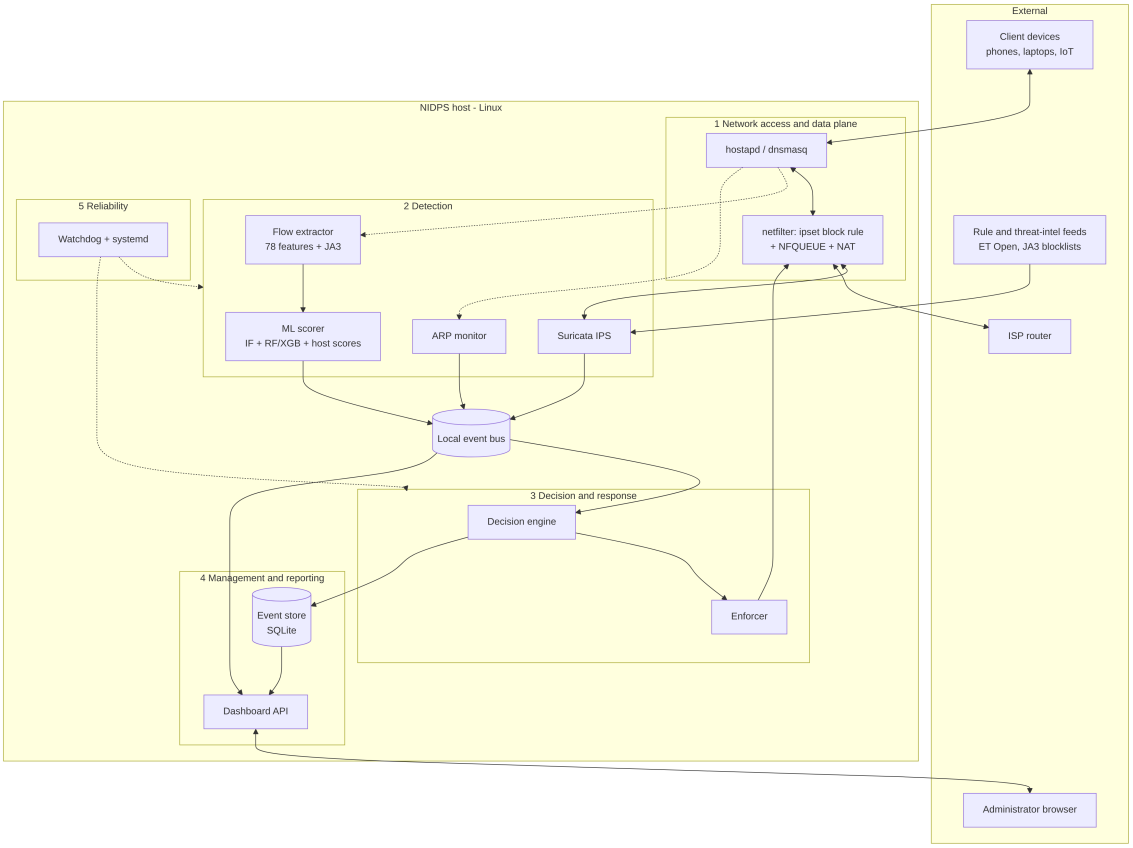


Figure 6.1: System architecture of the NIDPS.

Table 6.2: Subsystem responsibilities.

Subsystem	**Responsibility**	**Main components**
1. Network access and data plane	Provide Wi-Fi, addressing and routing; hold the inline queue; enforce blocks and failure mode in the kernel	hostapd, dnsmasq, netfilter (iptables, ipset, NFQUEUE, NAT)
2. Detection	Produce evidence: signature alerts, ARP alerts, flow records, anomaly scores, class probabilities	Suricata, ARP monitor, flow extractor, ML scorer
3. Decision and response	Turn evidence into verdicts and host risk; apply and expire blocks	Decision engine, enforcer
4. Management and reporting	Store events; expose live statistics, alerts and configuration to the admin	Event store, dashboard API and web UI, configuration manager
5. Reliability	Detect failed or overloaded components; restart them; apply failure mode; record recovery metrics	Watchdog, systemd units
Offline: training	Regenerate datasets, train and validate models, publish versioned model files	Dataset builder, trainer, model registry

The subsystems collaborate through two channels. The forwarding path is purely in the kernel and Suricata, with a single feedback loop from the enforcer into ipset. All other communication goes through the local event bus: detectors publish evidence, the decision engine consumes it and publishes verdicts, and the dashboard and event store subscribe to everything. Detectors therefore never call the decision engine directly, and a slow or crashed consumer never blocks a producer.

Rationale and rejected alternatives. Three decompositions were considered. A *single monolithic process* would be simplest to write, but one crash in the ML code would take down detection entirely, and memory would be shared by everything. A *distributed design* with a separate analysis server would allow larger models, but adds a second machine, network latency and cost, against the project's aim of one affordable box. The chosen *local service-oriented design* keeps everything on one host while isolating failures per service, which is what the failure-handling requirement needs.

6.3.1 Subsystem Architecture

This section describes each subsystem in more detail.

6.3.1.1 Network Access and Data Plane

This subsystem is mostly configuration of existing Linux components; the new parts are the netfilter rule layout and its management. Figure 6.2 traces one packet from a client to the internet.

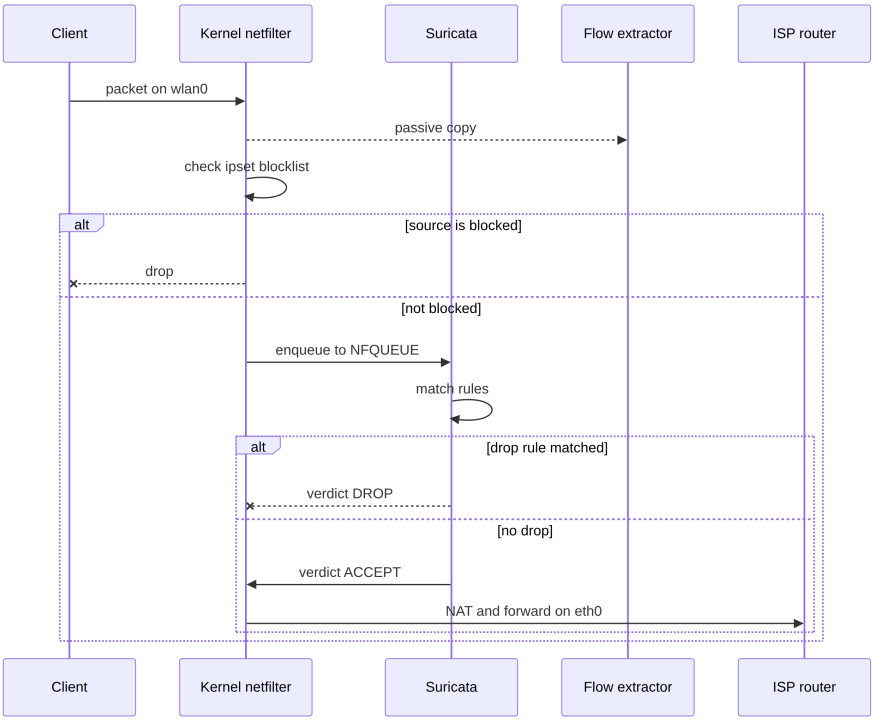


Figure 6.2: Packet path through the data plane.

The rule order matters: the ipset check comes before the queue, so packets from blocked hosts are discarded in the kernel and never cost Suricata any CPU. During a DDoS from a blocked source, this keeps the inspection engine free for other traffic.

6.3.1.2 Detection Subsystem

The detection subsystem has four producers. Suricata and the ARP monitor emit alerts directly. The flow extractor emits flow records, which the ML scorer enriches before publishing. Internally, the ML scorer is divided into three subcomponents:

- **Feature validator:** checks each flow record against the model's feature schema and rejects mismatched or malformed records.
- **Flow models:** the Isolation Forest and the RF/XGBoost classifier, both loaded once at start-up and applied to every flow record.
- **Host behaviour tracker:** keeps per-host state (outbound byte baselines, per-key connection timestamps, local JA3 counts) and computes the exfiltration and beaconing scores of Sections 5.5.3 and 5.5.4. Its state is bounded: entries for inactive hosts and keys expire.

6.3.1.3 Decision and Response Subsystem

The decision engine consumes all evidence from the bus, joins Suricata alerts with flow records by flow identifier and time, runs Algorithm 5.5 and maintains the host scores of Section 5.7.2. The enforcer receives BLOCK verdicts and runs Algorithm 5.6. The two are kept separate so that the component allowed to change firewall state is small, simple and easy to audit.

6.3.1.4 Reliability Subsystem

The watchdog receives heartbeats from every service and runs the state machine of Figure 5.3 for each. systemd provides process-level supervision (automatic restart, start order, resource limits), while the watchdog adds application-level health (queue depth, processing lag) and decides when to enter degraded mode. The kernel-level failure mode (queue bypass on or off) is set when the data plane starts, so it does not depend on the watchdog being alive.

6.3.1.5 Management and Reporting Subsystem

The dashboard API serves a web interface on the LAN side only. It provides live counters (flows per second, alerts and blocks per layer), an alert list with the evidence behind each verdict, the list of currently blocked hosts with a manual unblock action, host risk scores, component health, and configuration of thresholds, failure mode and allowlist. Configuration changes are validated, saved and published on the bus so running services pick them up without a restart.

6.3.1.6 Offline Training Pipeline

Figure 6.3 shows the offline pipeline and the hand-over of models to the device.

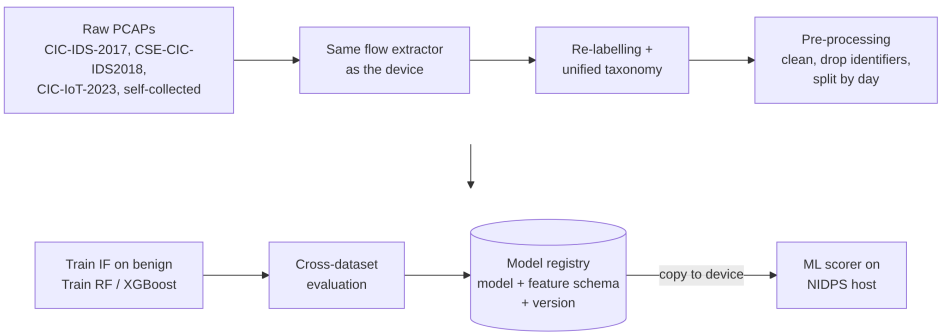


Figure 6.3: Offline training pipeline and model hand-over to the device.

Training runs on a workstation, not on the NIDPS host, because it needs far more memory than inference. The only hand-over to the device is a model package containing the model file, its feature schema and a version manifest.

6.4 Architectural Strategies

Each strategy below records the decision, the reasoning behind it and the alternatives that were rejected.

6.4.1 Layered Defence with a Short Fast Path

Decision: detection is split into a packet-level layer inside the forwarding path (Suricata) and flow-level layers outside it (ML), joined by a fusion step.
Reasoning: per-packet ML inference on every forwarded packet would exceed the CPU budget and add latency to every connection, violating the first goal of Section 6.2.3. Flow-level ML outside the path costs nothing in forwarding latency, and Suricata covers the attacks that must be stopped at the first packet.
Alternatives rejected: ML-only inline detection (too slow, and no immediate protection against known attacks); Suricata-only detection (misses zero-day and behavioural attacks, the gap identified in Chapter 1).

6.4.2 Metadata-Only Analysis of Encrypted Traffic

Decision: encrypted traffic is analysed only through JA3/JA3S fingerprints and flow statistics.
Reasoning: TLS interception would require a trusted root certificate on every client, fails against certificate pinning, breaks some IoT devices outright and would multiply CPU load. It would also expose all household traffic in plain text on one device, which is a serious privacy risk.
Alternatives rejected: full TLS interception (above); ignoring encrypted traffic (most C2 now uses TLS). **Known limitation:** browsers now randomise TLS extension order, which makes JA3 less stable for browser traffic. Because fingerprinting is isolated in the flow extractor, it can be switched to JA4, which sorts extensions before hashing, without changing other components.

6.4.3 Offline Training, Online Inference

Decision: models are trained on a workstation and deployed as read-only, versioned packages. The device only performs inference and threshold calibration.
Reasoning: training on hundreds of millions of flows needs far more memory than the device has. Online retraining on the device would also let an attacker gradually teach the model that malicious traffic is normal (model poisoning). Recalibrating only the threshold (Section 5.5.2) adapts to the local network with bounded risk.
Alternatives rejected: continuous online learning (poisoning risk, hardware cost); a fixed global threshold (false positives on real home traffic).

6.4.4 Local Services Connected by an Event Bus

Decision: each component runs as its own systemd service, and components exchange messages through a local publish/subscribe bus. Redis Streams is the proposed implementation.
Reasoning: service isolation means a crash in one component does not take down the others, and systemd can restart each independently, which is a precondition for the failure modes. The bus decouples producers from consumers. Redis Streams is lightweight, keeps a bounded backlog that a restarted consumer can resume from, and supports consumer groups so the dashboard and the decision engine read the same events independently.
Alternatives rejected: direct function calls in one process (no fault isolation); Apache Kafka (far too heavy for the hardware); plain Unix sockets between each pair of components (simple, but every new consumer needs new wiring, and messages are lost while a consumer restarts). ZeroMQ was a close alternative and remains a fallback if Redis proves too heavy.

6.4.5 Kernel-Enforced Blocking and Failure Semantics

Decision: both blocking (ipset with timeouts) and the failure mode (NFQUEUE bypass or not) are enforced by the kernel, not by user-space software.
Reasoning: user-space components can crash; the kernel keeps running. Timed ipset entries expire even if the enforcer dies, so a crash never leaves hosts blocked forever, and the queue-bypass flag applies the chosen mode even if the watchdog is down.
Alternatives rejected: iptables rules added and deleted one per blocked host (slow with many entries and leaves stale rules after a crash); a user-space proxy that drops traffic (adds latency and is itself a crash risk).

6.4.6 Choice of Languages and Products

Table 6.3 summarises the main product and language choices.

Table 6.3: Product and language choices.

Need	Choice	Main reason	Alternative considered
Signature IPS	Suricata	Multi-threaded, native NFQUEUE IPS mode, EVE JSON output, TLS/JA3 support, maintained ET Open rules	Snort 3 (similar capability; Suricata chosen for its JSON output and threading)
Services and ML	Python 3	scikit-learn and XGBoost available natively; fast development	C++ (faster, but much slower to develop; reconsidered only if profiling shows Python is the bottleneck)
Models	scikit-learn Isolation Forest and Random Forest, XGBoost	Small, fast tree models suited to tabular flow data	Deep autoencoders and LSTMs (higher compute cost, no clear gain on flow features)
Event store	SQLite	Zero administration, single file, enough for one device	PostgreSQL (needs a server process)
Dashboard	Python web API with WebSocket push and a lightweight web front end	Live updates with little code; reuses the Python stack	Grafana (heavy and less suited to showing verdict evidence and controls)
Firewall	iptables + ipset	Stated in the proposal; well documented; ipset handles large blocklists efficiently	nftables sets (modern equivalent; migration path kept open)

6.4.7 Error Detection and Recovery

Errors are handled at three levels.

1. **Record level:** malformed flow records or alerts are rejected by the feature validator and counted, never passed to models.
2. **Component level:** heartbeats and systemd supervision detect crashes and hangs and restart the component (Section 5.9).
3. **System level:** the kernel-enforced failure mode decides what happens to traffic while a component is down. All three levels log to the event store, so the dashboard can show when the system was not fully protected.

6.4.8 Data Storage, Persistence and Retention

Only derived data is stored: alerts, verdicts, block records, host risk history, calibration state and configuration. Raw packets and payloads are never written to disk in normal operation. Flow records are kept for a short rolling window (for example, 7 days) for calibration and investigation, then deleted. This protects household privacy and keeps disk use bounded on a small device.

6.4.9 Future Extension

The architecture is designed so that the likely extensions require adding a component rather than changing existing ones: a new detector publishes to the bus; a new response (for example, notifying the owner's phone) subscribes to verdicts; a newer model or JA4 support is deployed as a new model package or extractor version.

6.5 Domain Model/Class Diagram

Figure 6.4 shows the main classes of the runtime system. The central entity is the Flow, which carries its features and optional TLS fingerprint through the pipeline. All detectors implement a common Detector interface and produce Evidence; the DecisionEngine turns Evidence into a Verdict, and the Enforcer turns BLOCK verdicts into BlockEntry records. The Detector interface is the main extension point: a new detection technique is added by implementing it and publishing its Evidence, without changes to the DecisionEngine. The Enforcer is the only class that changes firewall state, which keeps that responsibility in one small, auditable place.

6.6 Policies and Tactics

The following policies affect implementation details rather than the overall structure of the system.

6.6.1 Rule and Threshold Management

The ET Open ruleset is updated daily by a scheduled job, tested for syntax with Suricata's test mode, and loaded with a live rule reload so inspection never stops. If the test fails, the previous ruleset stays active. Custom rules live in a separate file under version control, so updates to ET rules never overwrite them. All numeric thresholds (rule thresholds, flow timeouts T_a and T_i, τ, θ values, Θ, block durations) are read from one configuration file with documented defaults.

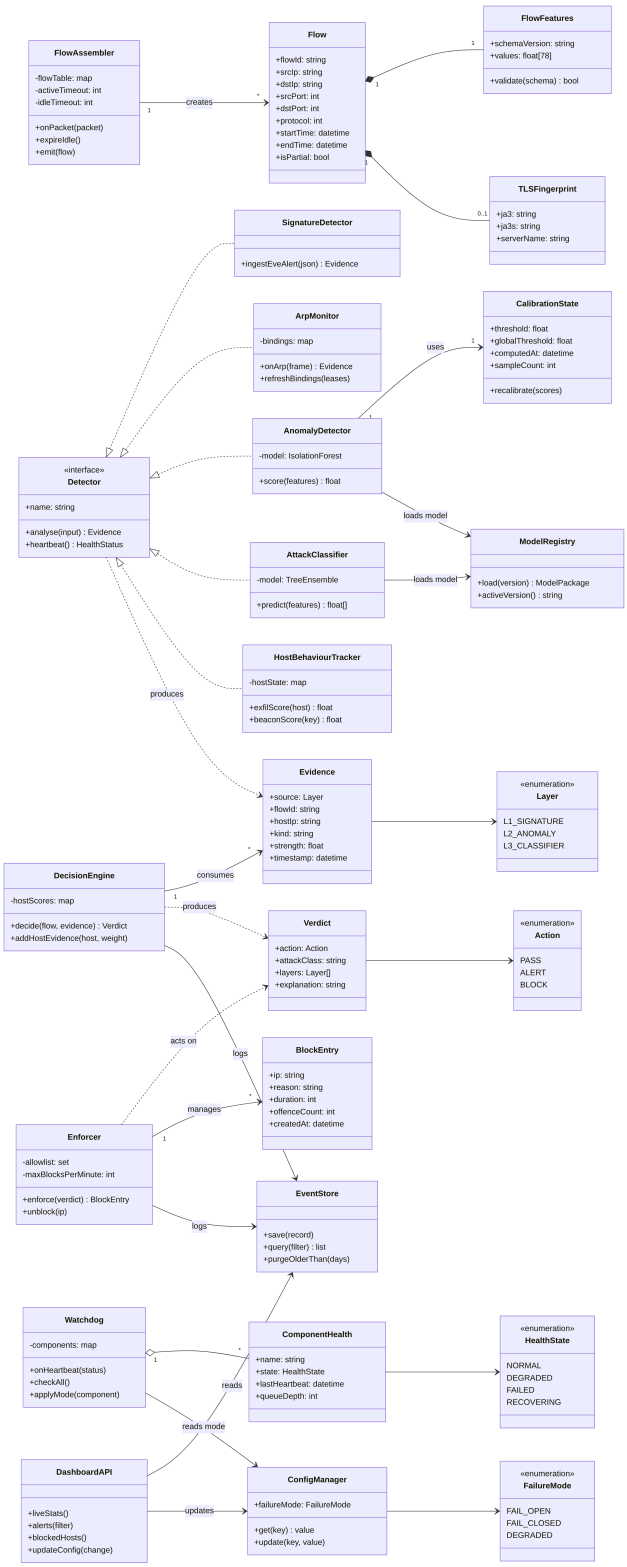


Figure 6.4: Class diagram of the NIDPS runtime.

6.6.2 Model Packaging and Versioning

Each model package holds the model file, the ordered feature list with the extractor version, the class list and the training date and metrics. At load time the ML scorer refuses any package whose feature list does not match the running extractor. XGBoost models are stored in its native JSON format rather than as Python pickles, since loading a pickle can execute arbitrary code. scikit-learn models use joblib files, loaded only from the registry directory, which is writable only by the deployment user. The previous package is kept so a bad model can be rolled back with one command.

6.6.3 Engineering Trade-offs

- **Detection speed vs. accuracy:** ML verdicts are per flow, not per packet (Section 5.7.3), accepting a short delay for ML-detected attacks in exchange for zero added forwarding latency.
- **False positives vs. missed attacks:** the calibration percentile p and the block thresholds are the explicit controls. Defaults lean towards fewer false positives, because wrong blocks on a home network cost the owner's trust quickly.
- **Availability vs. security:** chosen per deployment through the failure mode, with fail-open as the default for homes.

6.6.4 Coding Guidelines and Conventions

Python code follows PEP 8, uses type hints throughout and is checked with a linter and a type checker in continuous integration. Each service is a separate package with a single entry point. Messages on the event bus use a versioned JSON schema, so producers and consumers can be upgraded independently. Logs are structured JSON with a timestamp, component name and flow or host identifier, so events can be traced across services.

6.6.5 Requirements Traceability

Every goal from Chapter 2 is mapped to the components that implement it and the tests that verify it, in a traceability matrix kept in the repository and updated at every sprint review (Table 6.4).

Table 6.4: Requirements traceability matrix.

Proposal objective	**Implemented by**	**Verified by**
Transparent inline NIDPS between clients and ISP gateway	Data plane (Section 6.3.1)	Connectivity and latency tests (Table 5.7)
Suricata with ET ruleset for known attacks	SignatureDetector, custom rules (Section 5.4)	Live attack replay: SSH brute force, port scan, DNS spoofing
ML pipeline for anomaly detection and classification	AnomalyDetector, AttackClassifier, HostBehaviourTracker (Sections 5.5–5.6)	Detection metrics and live replay: C2 beaconing, exfiltration, DDoS, lateral movement
Cross-evaluation on three public datasets	Offline training pipeline (Figure 6.3)	Cross-dataset matrix (Table 5.6)
Automatic blocking via iptables	Enforcer (Section 5.8)	Time-to-block measurement; allowlist and rate-limit tests
Watchdog and configurable failure handling	Watchdog, systemd units (Section 5.9)	Fault-injection tests in each failure mode
Web dashboard	DashboardAPI (Section 6.3.1)	Functional UI tests; usability check with non-expert users

6.6.6 Testing Plan

1. **Unit tests** for each algorithm: flow assembly on crafted PCAPs, JA3 computation against published reference hashes, threshold calibration, fusion rules and the enforcer's allowlist, rate limit and back-off.
2. **Integration tests** that replay labelled PCAPs through the full pipeline on a test host and compare verdicts with labels.
3. **Live attack tests** on the isolated lab network, executing each in-scope attack from the attacker machine against our own test devices.
4. **Performance tests** for latency, throughput and resource use (Table 5.7).
5. **Fault-injection tests:** each service is killed or frozen under load in each failure mode, and recovery time and traffic impact are recorded.
6. **Soak test:** the system runs for at least one week on a real home network in alert-only mode to measure false positives before blocking is enabled.

6.6.7 Hierarchical Organisation of the Source Code

The source code is organised as follows:

```
nidps/
  dataplane/      hostapd, dnsmasq and netfilter configuration + setup scripts
  suricata/       suricata.yaml, custom.rules, rule-update job
  services/
    arpmon/       ARP consistency monitor
    flowextract/  flow assembler, feature computation, JA3/JA3S
    mlscorer/     feature validator, models, host behaviour tracker, calibration
    decision/     decision engine and host scores
    enforcer/     ipset management, allowlist, back-off
    watchdog/     heartbeat monitoring, state machine, failure modes
    dashboard/    web API and front end
  common/         bus client, message schemas, config loader, logging
  training/       dataset builder, labelling, training and evaluation scripts
  models/         model registry (versioned packages)
  deploy/         systemd unit files, install and upgrade scripts
  tests/          unit, integration, fault-injection tests and test PCAPs
  docs/           traceability matrix, design notes
```

6.6.8 Build and Deployment

The system is installed on a fresh Debian-based image by one installation script that installs system packages (Suricata, hostapd, dnsmasq, ipset, Redis), creates a Python virtual environment with pinned dependency versions, copies the configuration and the active model package, and enables the systemd units in dependency order: data plane first, then Suricata, then the remaining services. Upgrades replace one service at a time and restart only that service. Because every service is supervised and the failure mode is kernel-enforced, an upgrade never leaves the network without a defined behaviour.

6.6.9 Maintenance

Routine maintenance is automated: daily rule updates, weekly threshold recalibration (Algorithm 5.4), and log and flow-record purging beyond the retention window. Models are retrained offline when new datasets or significant new attack classes become available, or when the dashboard shows a rising alert rate that calibration does not explain. Component health history on the dashboard tells the owner when the system has been running in a degraded or failed state.

6.7 Conclusion

This chapter translated the methodology of Chapter 5 into a concrete design. The NIDPS is a single Linux host organised as five runtime subsystems (data plane, detection, decision and response, management and reporting, and reliability) plus an offline training pipeline. The main architectural decisions keep the forwarding path short, analyse encrypted traffic only through metadata, train offline while calibrating thresholds online, isolate components

as supervised services connected by an event bus, and let the kernel enforce both blocking and failure behaviour. The class diagram defines a common Detector interface that makes new detectors easy to add, while confining firewall changes to a single Enforcer. The policies on rule and model management, testing, traceability and deployment complete the design and connect each proposal objective to the component that delivers it and the test that verifies it.

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